12 Regenerative Endodontics

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CHAPTER OUTLINE

Overview of Regenerative Dentistry
Overview of Regenerative Endodontics
Preclinical Studies on Regenerative Endodontics
Stem Cells
Growth Factors/Morphogens
Morphogens
Scaffolds
Delivery System
Translational Studies
Summary of Basic Research on Regenerative Endodontics
Clinical Studies on Regenerative Endodontics
Clinical Procedures Related to Regenerative Endodontics
Overview of Clinical Regenerative Endodontic Procedures
Example of a Revascularization Protocol
Clinical Measures of Treatment Outcomes
Patient-Centered Outcomes
Clinician-Centered Outcomes
Scientist-Based Outcomes: Regeneration or Repair?
Summary

Overview of Regenerative Dentistry

Advancements in tissue engineering are dramatically changing medicine and dentistry. Tissue engineering is an interdisciplinary field that applies the principles of engineering and the life sciences toward restoring, maintaining, or replacing biologic function. It involves the interplay between stem cells, growth factors, and scaffolds (biologic matrices). It has become increasingly clear that the intentional manipulation of these three factors can lead to the regeneration of tissue function that would not otherwise take place if repair had taken place without intervention.[68] This relatively young field was first applied to medicine with many examples of regenerative medicine approaches used in the clinical practice.[175,222,306] Although the inclusion of tissue engineering in dentistry is more recent, it is also fundamentally changing the way clinicians are treating patients while providing a fertile research field that fosters future advancements and therapies.

Most of the history of dentistry is marked by the evolution of dental materials and techniques tailored to the replacement of lost or diseased tissues with inert materials. This prosthetic replacement of missing dental tissues has prevailed in dentistry since the primordial examples of dental treatments in ancient civilizations.[23,86,145,193,320] In contrast, the goal of regenerative dentistry is to induce biologic replacement of dental tissues and their supporting structures. The potential for regenerative dentistry is in large part due to advancements in biologic therapies that apply principles of tissue engineering with the spatial and temporal assembly of stem cells, growth factors, and scaffolds to achieve the functional regeneration of a missing tissue.

The foundations of regenerative endodontics date back to more than a century ago with the first attempts of vital pulp therapy with the goal of maintaining the physiologic function of an injured pulp.[71] Phillip Pfaff in 1756 performed the first reported pulp capping procedure. He concluded that the exposed injured pulp had the potential of repair if the tissue was further irritated with a caustic agent or heated instruments to promote the cauterization of the pulp stomps.[98,113] It was not until 1920 when Datwyler introduced zinc eugenol–based cement as a direct pulp capping agent.[125] Next, D.W. Herman introduced calcium hydroxide as a biocompatible agent to be used in both vital and nonvital therapies in 1921. In 1960 Professor Nygaard-Østby evaluated a revascularization method for reestablishing a pulp-dentin complex in permanent teeth with pulpal necrosis (see later).[220,221] Over the past several decades, the scope and clinical application of regenerative dental procedures have continuously advanced to now include guided tissue regeneration (GTR) or guided bone regeneration (GBR) procedures and distraction osteogenesis,[56,114] the application of platelet-rich plasma (PRP) for bone augmentation,[117] Emdogain for regeneration of periodontal tissues and pulp,[7,55] recombinant human bone morphogenic protein (rhBMP) for augmentation of bone,[61] and clinical trials on the use of fibroblast growth factor 2 (FGF-2) for periodontal tissue regeneration.[171] The potential of regenerative procedures in endodontics has been emphasized by elegant studies demonstrating the regeneration of pulp and dentin using scaffold materials and stem ce lls.[105,142,144,146,147,251] Thus regenerative dental procedures are emerging as a vital, evolving field of dental care, creating a paradigm shift in many dental specialties, including endodontics.[208] This chapter will review the current status of regenerative endodontic procedures (REPs) with an em-

phasis on biologic principles and the advantages and limitations of currently available clinical procedures.

OVERVIEW OF REGENERATIVE ENDODONTICS

The developing dentition is at risk for pulpal necrosis due to trauma, caries, and developmental dental anomalies such as dens evaginatus.[17-19,67,77,243,244,325] It is important to note that full radicular maturation occurs up to 3 years after the eruption of a tooth in the oral cavity,[204] and the loss of pulp vitality during this period arrests further root development, rendering these teeth susceptible to premature tooth loss. This untoward event in young patients with mixed dentition can have devastating consequences that include: altered maxillary and mandibular bone development, interferences with phonics, breathing, and mastication, and importantly, there is a severe detrimental psychosocial effect on young patients.[158,287]

Immature teeth diagnosed with pulp necrosis have been traditionally treated with apexification procedures using either long-term calcium hydroxide treatment[65,66] or immediate placement of a mineral trioxide aggregate (MTA) apical plug.[315] Although these treatments often result in the resolution of signs and symptoms of pathosis, they provide little to no benefit for continued root development.[39] The process of root maturation includes root elongation, apical closure, and further root thickening (Fig. 12.1). This process is mediated through epithelial-mesenchymal interactions between the Hertwig root sheath (HERS) and the apical papilla.[317] Throughout this process, the pulp-dentin complex is formed at the expense of the apical papilla as stem cells migrate and differentiate forming odontoblasts and fibroblasts, reducing the apical papilla size as the root matures (see Fig. 12.1). Unfortunately, this elaborate developmental process can be disrupted by untoward events such as trauma and infection. Thus immature teeth treated with these procedures are considered in a state of "arrested development," and no further root growth, normal pulpal nociception, and immune defense should be expected.

REPs have been defined as biologically based procedures designed to replace damaged structures such as dentin, root structures, and cells of the pulp-dentin complex.[208] This new treatment modality has emerged as a treatment alternative for these teeth that, in addition to healing of apical periodontitis, aims to promote normal pulpal physiologic functions. These include continued root development, immune competency,[24] and normal nociception, as seen in some published cases.[77] Thus the ultimate goal of these procedures is to regenerate the components and normal function of the pulp-dentin complex to promote tooth longevity and function.

Regenerative endodontics has its foundation on the seminal work by Dr. Nygaard-Østby more than 50 years ago.[226] He hypothesized that a blood clot could be the first step in the healing of a damaged dental pulp, similar to the role of the blood clot in the healing process observed in other areas (e.g., alveolar bone following extraction). To test the hypothesis that the presence of a blood clot within a root canal system promotes healing, mature teeth diagnosed with either vital or necrotic pulp received pulp space débridement followed by foraminal enlargement, medicament dressing for the necrotic cases, evoked intracanal bleeding, and a kloroperka obturation placed coronal to the formed blood clot. Patients ($n = 17$) were followed for various time periods (17 days to 3.5 years), and then the treated tooth was extracted and the newly formed tissues were histologically examined. The outcomes were similar for all teeth: (1) resolution of symptoms of inflammation related to foraminal enlargement and over-instrumentation in as early as 17 days, (2) resolution of signs and symptoms of pathosis for the necrotic cases, and in certain cases, (3) radiographic evidence of apical closure. For the histologic analysis, it was observed that there was ingrowth of connective tissue into the canal space and varied levels of mineralized tissue found along the canal walls as well as "islands" of mineralized tissue embedded within the

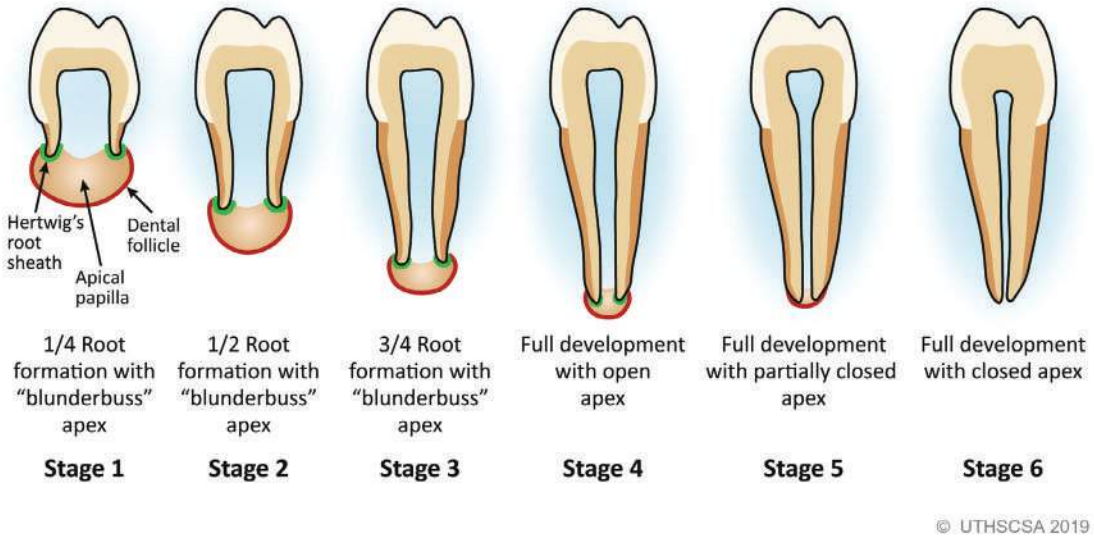


Fig. 12.1 Schematic drawing demonstrating the different stages of root development and the subsequent reduction of the apical papilla and Hertwig root sheath.

newly formed tissue (see Fig. 12.1). Because dental pulp is a type of connective tissue with a rich supply of fibroblasts, this general finding was quite promising. However, the inclusion of undesired cell types (e.g., cementoblasts) and the lack of desired cell types (e.g., odontoblasts) indicate that this protocol did not lead to complete histologic regeneration of dental pulp. Despite its shortcomings, this pioneer study laid the foundation for the subsequent studies in the field of regenerative endodontics.

In 1966 a study was published reporting that disinfection could be established primarily with interappointment medication with polyantibiotic mixes (three different formulations used in five cases).[245] The investigators did not purposely evoke intracanal bleeding in this study but instrumented canals short of what they believed to be vital tissue determined by visualization of tissue and pain sensation upon instrumentation. Resolution of signs and symptoms of disease and continued root development was seen for all reported cases. It represented the first reported case where polyantibiotic pastes were used in immature necrotic teeth for disinfection and to promote root development. Five years later, another study was published that included the use of antibiotics in the disinfection protocol and the intentional promotion of intracanal bleeding.[221] Resolution of symptoms and continued root development was a common finding. However, histology of the extracted teeth demonstrated that connective tissue was formed in 28 out of 35 teeth, whereas cellular cementum was formed in 18 out of 35 teeth. Again, these protocols generated acceptable clinical outcomes (e.g., healing of apical periodontitis, lack of symptoms) with only partial evidence of dental pulp phenotype. Collectively, these findings laid the foundation for contemporary regenerative endodontics, demonstrating that repair could take place following root canal disinfection in immature teeth.

The emergence of the field of regenerative endodontics was catalyzed in the early 2000s with the publication of two remarkable case reports.[27,149] Since then, there has been an exponential increase in published cases reporting unprecedented clinical outcomes such as resolution of signs and symptoms of apical periodontitis, continued root development, and, in certain cases, normal nociceptive responses to vitality testing.[77,130,173] Among the many published cases, there are two randomized clinical trials[185,211] that, despite limited sample sizes, support the hypothesis that patients with otherwise limited treatment options benefit from these procedures.[297] Importantly, the field of regenerative endodontics has seen a dramatic increase in knowledge gained from translational basic science studies evaluating the interplay of the tissue-engineering components (stem cells, growth factors, and scaffolds) applied to the clinical need and challenges.

Preclinical Studies on Regenerative Endodontics

Applying the principles of tissue engineering to the development of REPs requires research on the correct spatial assembly of distinct stem cells, growth factors/morphogens, and scaffolds to form a functional pulp-dentin complex.[131,142,144,176] In this section, we will review each of these critical components in turn.

STEM CELLS

Stem cells are defined as a distinct subpopulation of undifferentiated cells with the ability of self-renewal and differentiation into cell lineages of their tissue of origin.[203] They can be classified as pluripotent or multipotent cells. Pluripotent stem cells have the capacity of becoming specialized cells belonging to all three germ layers. Embryonic stem cells are the best example of pluripotent cells. There is a significant body of research on embryonic stem cells, but ethical, legal, and medical (tissue-rejection) issues can render these cell types unsuitable for clinical applications.[208] True pluripotent stem cells can be found only in the developing embryo, and the harvesting of these cells requires destruction of the embryo, hence the legal and ethical concerns with such practice. Yamanka and colleagues reported the groundbreaking finding that somatic cells can be transformed into pluripotent stem cells, namely induced pluripotent stem cells (iPSCs).[225] The use of iPSCs does not have the same legal and ethical concerns as the use of embryonic stem cells, but there are still concerns regarding their inherent pluripotent capacity and lack of proliferative control. These cells tend to form teratomas after implantation into a host, a true testament of their high proliferative and differentiation capacities, but making their use in clinical applications challenging, requiring that differentiation into single phenotypically differentiated cells occurs in vitro prior to implantation.[165] On the other hand, all adult mesenchymal stem cells (MSCs) are more restricted in their capacity to differentiate, forming tissues only of mesenchymal origin, and therefore are classified as multipotent.[50] These cells can be found compartmentalized within tissues in "stem cell niches." The mesenchymal tissues (e.g., bone, dental pulp, periodontal ligament) appear to have an enriched population of adult stem cells.[50] These cells were first found in bone marrow decades ago and characterized as self-renewing and plastic-adherent and formed cell colonies with a fibroblastic appearance.[100,101] They were initially called stromal stem cells but later received the now widely accepted terminology "mesenchymal stem cells."[50] It is worth mentioning this terminology is not unanimous, and the characteristics or criteria that define a "stem cell" remain controversial.[51] Nonetheless, most undifferentiated progenitor cells found in the orofacial region are considered MSCs.[83]

Different populations of adult stem cells have been identified in tissue compartments in the oral region. These include stem cells of the apical papilla (SCAPs),[279] inflammatory periapical progenitor cells (iPAPCs),[184] dental follicle stem cells (DFSCs),[205] dental pulp stem cells (DPSCs),[124] periodontal ligament stem cells (PDLSCs),[261] bone marrow stem cells (BMSCs),[100] tooth germ progenitor cells (TGPCs),[285] salivary gland stem cells (SGSCs),[260] stem cells from human exfoliated deciduous teeth (SHEDs),[201] oral epithelial stem cells (OESCs), gingiva-derived mesenchymal stem cells (GMSCs),[156] and periosteal-derived stem cells (PSCs) (Fig. 12.2).[326] Although stem cells have been identified in most oral tissues, the stem cells more likely to be involved in REPs are localized around the periapical region. These include SCAPs, PDLSCs, BMSCs, iPAPCs, and DPSCs (if vital pulp is still present apically).

The apical papilla and its residing stem cells (SCAPs) were first characterized in 2006.[279] The apical papilla (Fig. 12.3)

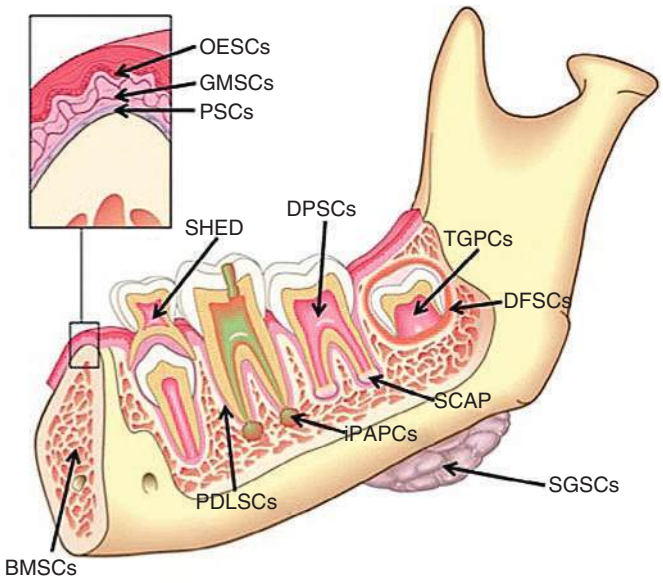


Fig. 12.2 Schematic drawing illustrating potential sources of postnatal stem cells in the oral environment. Cell types include tooth germ progenitor cells *(TGPCs)*, dental follicle stem cells *(DFSCs)*, salivary gland stem cells *(SGSCs)*, stem cells of the apical papilla *(SCAPs)*, dental pulp stem cells *(DPSCs)*, inflamed periapical progenitor cells *(iPAPCs)*, stem cells from human exfoliated deciduous teeth *(SHEDs)*, periodontal ligament stem cells *(PDLSCs)*, bone marrow stem cells *(BMSCs)*, and, as illustrated in the inset, oral epithelial stem cells *(OESCs)*, gingiva-derived mesenchymal stem cells *(GMSCs)*, and periosteal stem cells *(PSCs)*. (From Hargreaves KM, Diogenes A, Teixeira FB: Treatment options: biological basis of regenerative endodontic procedures, *J Endod* 39:s30–s43, 2013.)

is a dense reservoir of undifferentiated MSCs with great proliferative and odontogenic differentiation capacity and are directly involved in root formation during development.[140,246,317] Furthermore, the close proximity of the apical papilla to the apices of teeth in continuum with the root canal space makes this rich source of stem cells readily available for regenerative endodontic therapeutics through autologous cell transfer without the necessity to harvest and manipulate these cells exogenously. The iPAPCs represent another important potential source of stem cells for regenerative endodontics in teeth with well-established apical periodontitis.[58,184] These cells appear to be largely localized in the vasculature within apical granulomatous tissues. Lastly, PDLSCs and BMSCs should be also considered as stem cells sources for regenerative procedures because the action of mechanical disruption of the apical tissue (evoked bleeding) could also trigger the release of these cells, albeit their relative abundance is believed to be significantly less than SCAPs and iPAPCs.

In 2011 a study was conducted to evaluate the presence of MSCs following the evoked-bleeding step in regenerative procedures.[191] It was found that there is a substantial influx of MSCs into root canals during regenerative procedures resulting in an increase greater than 700-fold in the expression of MSC markers (Fig. 12.4). In addition, the cells could be harvested from clinical samples and examined under confocal microscopy (Fig. 12.5). This was the first demonstration that REPs are stem cell–based therapies.[77] Although this study did not evaluate whether the MSCs detected in REPs are derived from the apical papilla, it was assumed that these cells were SCAPs because the evoked-bleeding step lacerated the apical papilla. However, these MSCs are a heterogeneous population of cells that could come from any of the periradicular tissues after the mechanical step of evoking bleeding into the root canal system.

The delivery of substantial concentrations of MSCs into the root canal space, despite advanced apical periodontitis or abscess, points to an impressive survival capacity of these cells. In these clinical presentations, low oxygen tension, low pH, and high concentration of endotoxins and inflammatory mediators are expected. Indeed, the finding of high concentrations of the immune cell marker CD14 in those clinical samples indicates that there was still a substantial

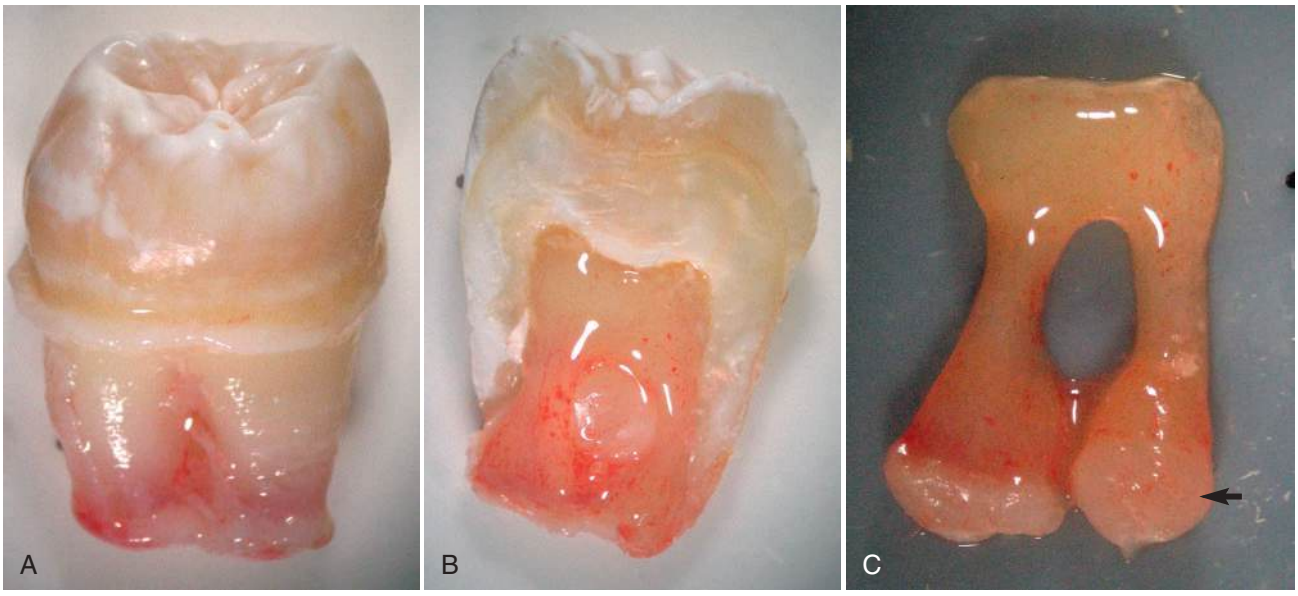


Fig. 12.3 Dissection of an immature permanent tooth indicating the extent of the apical papilla (**A–C**). Note that this structure is likely lacerated during the evoked bleeding step of revascularization cases and thus cells from this structure, including mesenchymal stem cells of the apical papilla (SCAPs), are likely to be delivered into the root canal space. *Arrow* in **C** denotes junction of apical papilla and dental pulp. (Courtesy Dr. Michael Henry.)

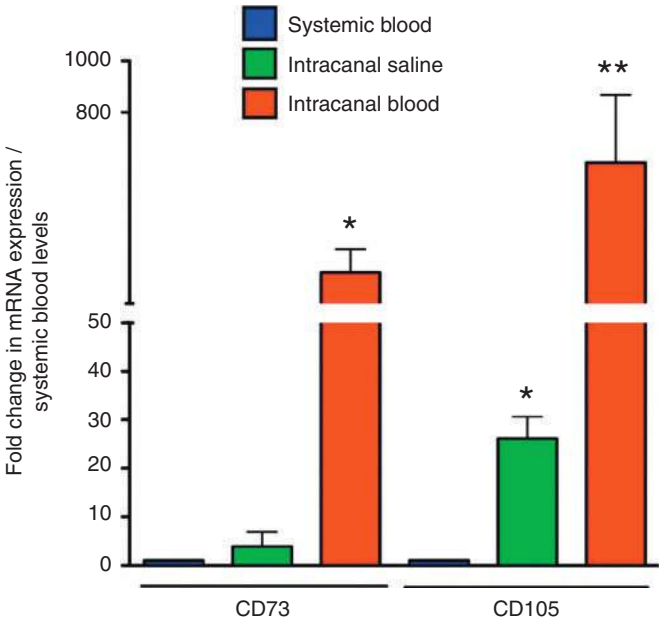


Fig. 12.4 Evoked-bleeding step in endodontic regenerative procedures in immature teeth with open apices leads to significant increase in expression of undifferentiated mesenchymal stem cell markers in the root canal space. Systemic blood, saline irrigation, and intracanal blood samples were collected during second visit of regenerative procedures. Real-time RT-PCR was performed by using RNA isolated from each sample as template, with validated specific primers for target genes and 18S ribosomal RNA endogenous control. Expression of mesenchymal stem cell markers CD73 and CD105 was upregulated after the evoked-bleeding step in regenerative procedures. Data were normalized to the housekeeping gene 18S levels and presented as mean ± standard deviation fold increase in relation to systemic blood levels for each gene and analyzed with one-way analysis of variance with Bonferroni post hoc test ($n = 8$; $^*P < .05$; $^{**}P < .01$; n.s., not statistically significant). (From Lovelace TW, Henry MA, Hargreaves KM, Diogenes A: Evaluation of the delivery of mesenchymal stem cells into the root canal space of necrotic immature teeth after clinical regenerative endodontic procedure, *J Endod* 37:133–138, 2011.)

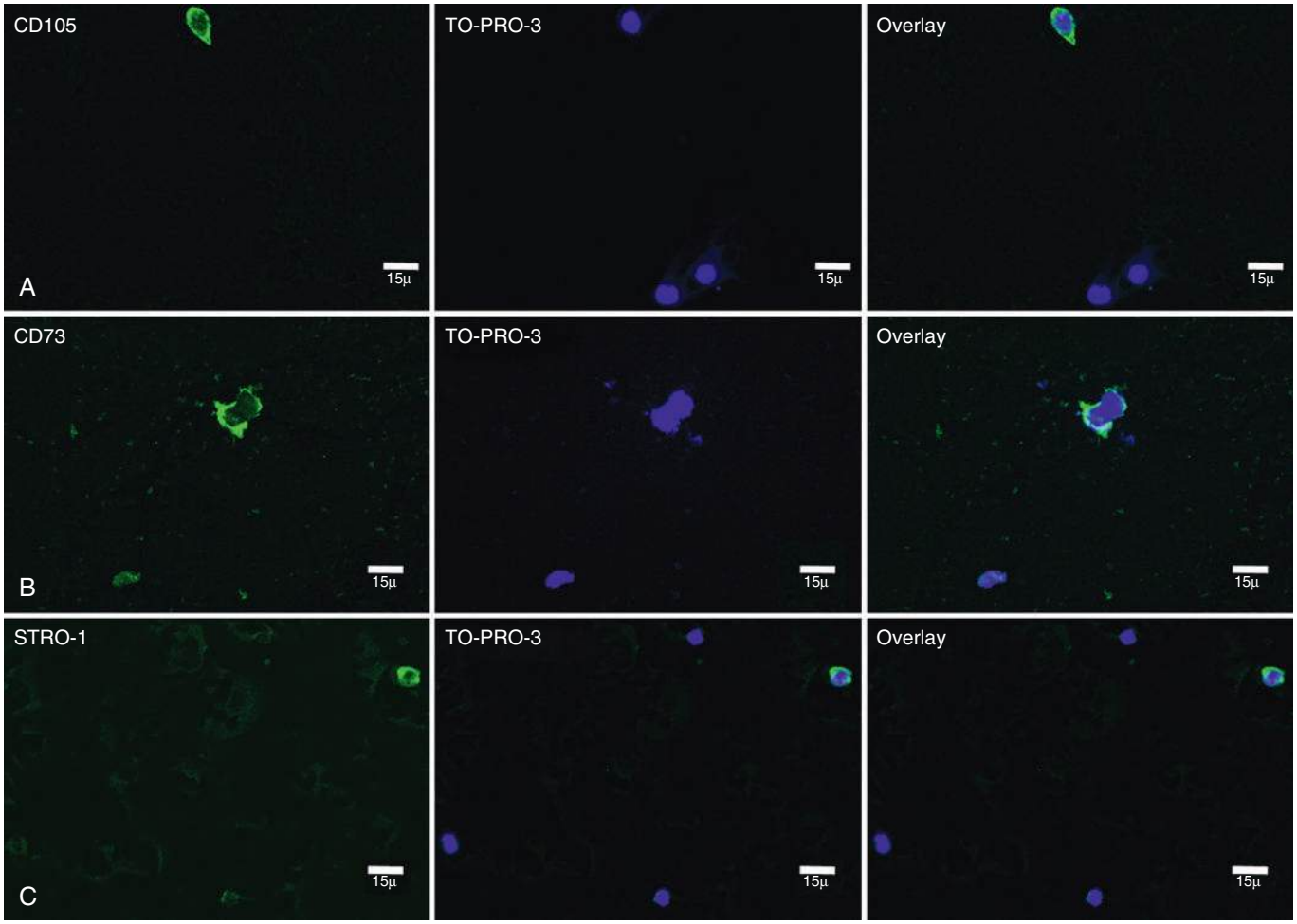


Fig. 12.5 Mesenchymal stem cells were delivered into root canal spaces during regenerative procedures in immature teeth with open apices. Cells collected from intracanal blood samples after the evoked-bleeding step or from systemic blood were stained with antibodies against CD105, CD73, or Stro-1 and evaluated with a laser-scanning confocal microscope. (**A**) Cells in intracanal blood samples collected after evoked-bleeding step showed expression of mesenchymal stem cell marker and CD105 (*green* in **A**), CD73 (*green* in **B**), and STRO-1 (*green* in **C**), whereas nuclei appear blue as stained with TO-PRO-3. (From Lovelace TW, Henry MA, Hargreaves KM, Diogenes A: Evaluation of the delivery of mesenchymal stem cells into the root canal space of necrotic immature teeth after clinical regenerative endodontic procedure, *J Endod* 37:133–138, 2011.)

chronic inflammatory exudate present at the apical region of those teeth. These findings raise the question of how MSCs such as SCAPs can survive during apical periodontitis where a complex microflora, an array of inflammatory mediators, immune cells, and presumably low oxygen tension are commonly encountered. The biologic reason for this apparent resilient survival, despite challenging conditions, may be explained by the relatively low density of blood vessels in the apical papilla in comparison with the adjacent dental pulp, whereas the dental follicle surrounding the apical papilla is highly vascularized and may act as a capillary bed to supply nutrients to SCAPs.[77] Indeed, the apical papilla was found to remain vital despite complete pulpal necrosis and advanced apical periodontitis in animal model of endodontic infection in rodents[290] and dogs.[229] Further evidence of this survival capacity was shown in a clinical case when an apical papilla was harvested through the canal space with a Hedstrom file. This tissue was processed for immunohistochemistry, and cells were cultured and found to be multipotent and expressing all recognized marked of MSCs.[59] The cells cultured from the apical papilla of that patient with a history of apical abscess due to pulp necrosis for more than 1 year showed greater mineralization potential than apical papilla cells cultured from healthy extracted third molars.[59] Interestingly, it has been demonstrated that hypoxic environments enhance the proliferation, survival, and angiogenic potential of dental stem cells[4,21,84,305] and that similar enhancing effects were observed when dental stem cells were exposed to bacterial by-products or antigens.[84,308] Thus it appears that SCAPs and surrounding stem cells are equipped to survive and maintain their potential for differentiation in adverse conditions such as apical periodontitis and apical abscesses. Nonetheless, stem cells delivered into root canals after bleeding is evoked from the periradicular tissues are likely from various apical sources or niches, and not restricted to the apical papilla.

A clinical study in 2014 evaluated the intracanal transfer of undifferentiated MSCs from the apical region in adult patients through evoked bleeding.[58] A substantial increase in MSC markers could be detected in intracanal blood samples compared with systemic blood using quantitative real-time polymerase chain reaction (qRT-PCR). In addition, cells were cultured from the intracanal blood sample from tooth #8 in a 35-year-old female patient diagnosed with pulp necrosis and symptomatic apical periodontitis. The cells cultured were adherent to plastic, formed single colonies, coexpressed MSC markers, had high proliferative rates, and differentiated into a mineralizing phenotype.[58] This study supports that apical lesions contain a substantial number of undifferentiated cells and that these cells could participate in an intracanal regenerative or reparative process. Indeed, these procedures have been attempted in adults with fully formed teeth and published as case report[232,318,331] and case series.[252] Although significant root development is not expected in mature roots, it can be argued that regaining nociception and immune defense, a tooth regains surveillance and reparative functions, respectively. However, further evidence is needed to compare the predictability and the long-term success and survival of mature teeth treated with a regenerative approach versus conventional root canal therapy. Regardless of the clinical outcome of current techniques, evidence strongly supports the presence of apical undifferentiated cells in mature teeth in adults, opening the possibility for future biologic-based therapies for all patients and not only children with immature teeth.

The dental pulp can be viewed as a core of innervated and vascularized loose connective tissue surrounded by a layer of odontoblasts. The major cell type of this core region is the fibroblast. Together with blood vessels, lymphatics, and neurons, this core tissue is embedded in an extracellular matrix consisting of collagen and other fiber types (see also Chapter 13). DPSCs can be found throughout the dental pulp but are known to accumulate in the perivascular region and the cell-rich zone of Hohl adjacent to the odontoblastic layer.[96,262] Thus DPSCs from both sources are believed to be active participants in the process of reparative dentinogenesis. These cells are recruited to the site of injury following a gradient of chemotactic agents released by resident immune cells and from the damaged dentin.[1,270] The reparative dentin formed by these cells is distinct from the primary, secondary, and reactionary dentin that has been lost.[8,324] It is often called "osteodentin" when found to be disorganized and atubular and having cellular inclusions.[302] This process of cellular repair is known to be enhanced by bioactive materials (e.g., MTA and Biodentine).[168] These materials are known to increase the inherent mineralizing potential of the dental pulp when used in both indirect and direct pulp applications. These biologic events form the basis of contemporary vital pulp therapies that include direct pulp capping and pulpotomies with well-documented high success rates.[137,188,266] However, the physiologic process of tertiary dentinogenesis requires a vital pulp and the resolution of the etiology (e.g., caries or trauma). This process becomes disrupted when the pulp succumbs to injury, resulting in liquefaction necrosis of the dental pulp. Regeneration or repair in this case is only possible with the recruitment or delivery of autologous stem cells to the canal space following adequate disinfection.[77]

Odontoblasts are one of the most specialized cells of the pulp-dentin complex with dentinogenic, immunogenic, and possibly sensorial functions.[40,82,85] Identification of odontoblasts in the intact pulp-dentin complex is easily achieved based on their location and their distinct morphologic characteristics (i.e., columnar polarized cell body with cellular projections into the dentinal tubules). However, identification and characterization of an odontoblast-like cell are far more challenging. One of the main reasons for this is that these cells lack a primary odontoblast morphology and unique markers that could be used for identification.[139] Indeed, many markers used for the identification of odontoblast-like cells are also expressed in other mineralizing cell types such as osteoblasts. For example, both odontoblasts-like cells and osteoblasts are similar in the formation of mineralized nodules and in the expression of several proteins such as dentin sialoprotein (DSP), although DSP levels are nearly 400 times greater in odontoblasts than osteoblasts.[309] Measuring only one or two characteristics of a cell might not conclusively identify whether the cell is a true odontoblast. Even among odontoblasts, the phenotype varies in cells located in the apical (squamous shape) versus coronal (tall columnar) pulpal tissue. Importantly, molecular studies have identified many of the genes selectively expressed in odontoblasts.[189,227,228] Lastly, an intermediate filament protein called

"nestin" has been shown to be preferentially expressed in odontoblasts or odontoblast-like cells when in the active secretory function. Nestin expression could be used in conjunction with other markers to better identify odontoblast-like cells.[2,3] This knowledge is expected to aid future studies characterizing the conditions necessary for mesenchymal cells of multiple origins to differentiate into odontoblast-like cells. It is likely that definitive cellular identification depends upon both the morphology of the cell and an assessment of the expression of multiple genes.

At least five different types of postnatal MSCs, in addition to DPSCs, have been reported to differentiate into odontoblast-like cells, including SHEDs,[201] SCAPs,[140] iPAPCs,[184] dental follicle progenitor cells (DFPCs),[205] and bone marrow mesenchymal stem cells (BMMSCs).[26] As stated previously, overinstrumentation into the periradicular tissues followed by bleeding into the canal space results in a robust influx of cells with MSC markers in fully mature teeth, similarly to that seen in immature teeth.[58,191] Thus it appears that MSCs from the apical region can be delivered into the root canal spaces in both immature and mature teeth. However, there is growing evidence that MSCs have decreased proliferative and differentiation potential with aging.[166,215] Further research is required to elucidate the age limit for the use of autologous dental-derived stem cells, but these findings suggest that regenerative procedures may be applicable to mature fully formed teeth in adults. Thus it is recognized that more research and development are required to make regenerative procedures more predictable for immature teeth and that these procedures may transition to be applicable to fully formed teeth.

GROWTH FACTORS/MORPHOGENS

Dentin is composed of collagen fibers (90%, collagen type I) and noncollagenous matrix molecules (proteoglycans, phosphoproteins, and phospholipids). The collagen fibers act as a grid or matrix, and this structure behaves as a scaffold upon which mineralization can occur. Dentin phosphoprotein (DPP) and DSP are the most abundant dentin-specific proteins among the noncollagenous proteins of organic matrix.[46] DSP resembles other sialoproteins such as bone sialoprotein, but its precise function is still unclear; it may have a role in matrix mineralization.[126] Both DSP and DPP are a part of the small integrin-binding, ligand N-linked glycoproteins (SIBLINGs) which include dentin matrix acidic phosphoprotein 1 (DMP-1), bone sialoprotein, osteopontin, osteocalcin, and osteonectin. These proteins are only a small part of the whole cocktail of noncollagenous proteins which form components of the dentin.[119]

Research on dentin structure and composition has highlighted that the matrix contains some components that may be important in regulating tissue due to their bioactive properties. Based on this, dentin is currently considered a reservoir of growth factors and cytokines.[32,270,274–276] These growth factors/cytokines are secreted by the odontoblasts during primary dentinogenesis, becoming sequestered and trapped or "fossilized" into the dentin matrix after biomineralization (Table 12.1). However, they may become solubilized by demineralization of the matrix, by bacterial acid (caries decay), chemical treatment (ethylenediaminetetraacetic acid [EDTA] rinsing solution, calcium hydroxide or acid etching for bonded restorations), or by restorative materials such as MTA and Biodentine (Fig. 12.6).[272,292]

Table 12.1 List of Growth Factors Found into the Mineralized Dentin Matrix

Growth Factors in Dentin Matrix	
Transforming growth factor beta-1 (TGF-β1)	Cassidy et al. (1997)[53]
Transforming growth factor beta-2 (TGF-β2)	Cassidy et al. (1997)[53]
Transforming growth factor beta-3 (TGF-β3)	Cassidy et al. (1997)[53]
Bone morphogenic protein 2 (BMP-2)	Thomadakis et al. (1999)[289a]
Bone morphogenic protein 4 (BMP-4)	About et al. (2000)[3]
Bone morphogenic protein 7 (BMP-7)	Thomadakis et al. (1999)[289a]
Insulin-like growth factor-1 (IGF-I)	Finkelman et al. (1990)[95]
Insulin-like growth factor-2 (IGF-II)	Finkelman et al. (1990)[95]
Hepatocyte growth factor (HGF)	Tomson et al. (2013)[292a]
Vascular endothelial growth factor (VEGF)	Roberts-Clark and Smith (2000)[242a]
Adrenomedullin (ADM)	Musson et al. (2010)[208a]

Other growth factors and their receptors have been shown to be present at the enamel organ–dental papilla interface by immunohistochemistry and in situ hybridization during tooth development and have been implicated in odontoblast differentiation:

- Growth hormone (GH) plays a paracrine and/or autocrine role in dental development.[329]
- Insulin-like growth factor beta-1 (IGF-β1) and beta-2 (IFG-β2)[31,157]
- Transforming growth factor beta-1 (TGF-β1), TGF-β2, and TGF-β3[69,288] and bone morphogenic protein 2 (BMP-2), BMP-4, and BMP-6[303] play a role in the polarization and the differentiation of odontoblasts.[31] Notably in adult pulp, TGF-β1 plays an important role in the regulation of the inflammatory response and tissue regenerative processes.[183]

The dental pulp has well-recognized regenerative potential observed in the process of reparative dentinogenesis.[207,271] In this process, dentin-derived growth factors are believed to play a key role to be deciphered into the regulation of progenitor cell recruitment, cell proliferation, and differentiation of new dentin-secreting cells.[183,270] Indeed, the differentiation of new odontoblast-like cells has also been reported following pulp capping with fibroblast growth factor (FGF), TGF-β1 (Li et al. 2014), and BMP-7.[118,183]

The sequestration of these growth factors in the dentin matrix and their subsequent "fossilization" during the mineralization process appear key to the pulp healing process where their release from the matrix may be responsible for various signaling events. These growth factors are extremely potent and have a variety of cell signaling properties. However, their precise localization in the dentin[273] and their various biologic roles remain to be elucidated.

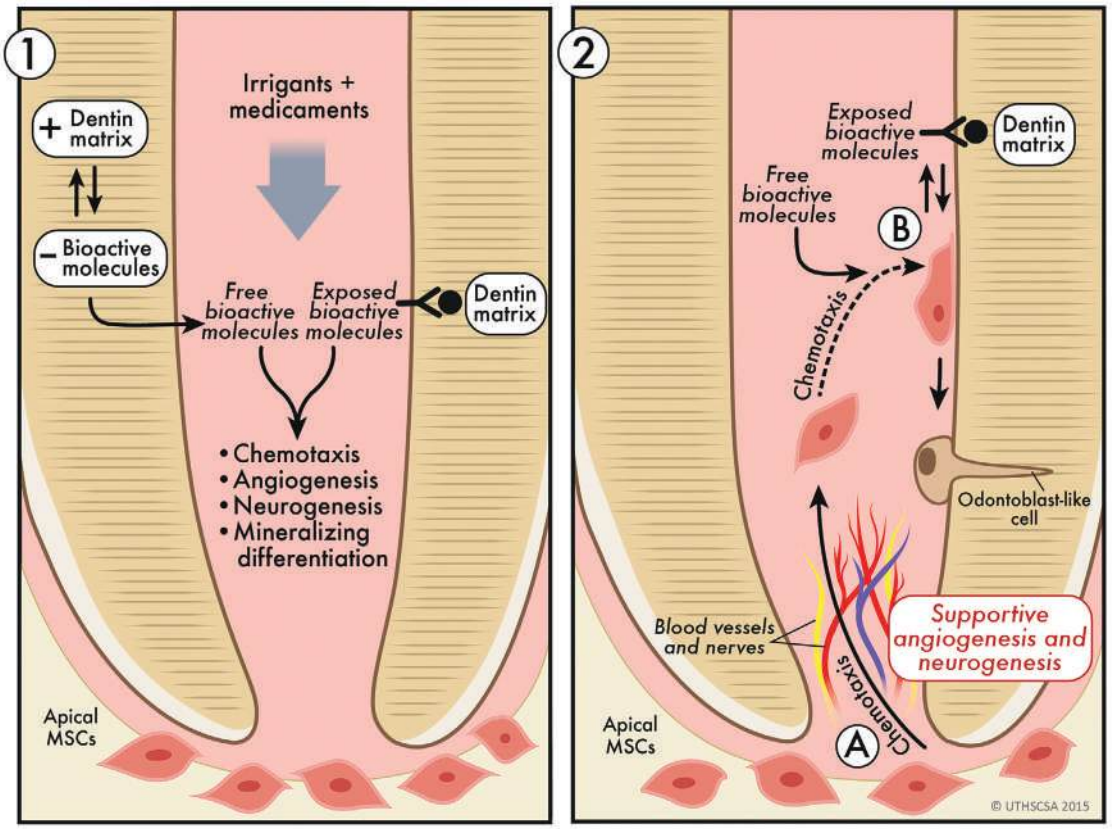


Fig. 12.6 Schematic illustrating the potential actions of irrigants and medicaments in the release and/or exposure of bioactive molecules sequestered in dentin *(1)* and their influences on regenerative events, including chemotaxis of apical undifferentiated cells *(A)*, odontoblast-like cell differentiation *(B)*, mineralization, angiogenesis, and neurogenesis *(2)*. (From Smith AJ, Duncan HF, Diogenes A, Simon S, Cooper PR. Exploiting the bioactive properties of the dentin-pulp complex in regenerative endodontics, *J Endod* 42(1):47–56, 2016.)

It is possible to imagine opportunities for therapeutic stimulation, inducing a targeted release of these proteins. For example, treatment of dentin with EDTA solution has been shown to dissolve the mineral phase, liberating growth factors that orchestrate the stimulation of progenitors or stem cell differentiation.[104,105,110,270,274,276] Etching with orthophosphoric acid, used for conditioning the dentin in bonding procedures, also promotes demineralization of the dentin and liberation of biologic factors.[81,93] For a long time, calcium hydroxide has been used as a protective lining, especially beneath amalgams fillings, or as a canal disinfection medication. It has been shown to have the ability to release bioactive components from the dentin, including growth factors.[123] Unlike dentin-etching acids which have only brief contact with the dentin, calcium hydroxide remains in place beneath restorations or in canals, allowing for a gentle and continued dissolution, thus releasing growth factors; its action is prolonged and potentially controllable depending on the form of the product. Lastly, calcium hydroxide, a by-product of the use of MTA and Biodentine, appears to underlie release of bioactive dentin-derived growth factors by these two bioactive materials.[292] Thus clinicians may take advantage of potent growth factors stored within dentin with the use of chemical treatments and materials that promote the release of these factors (see Fig. 12.6).

Morphogens

It is also important to keep in mind that a second level of regulation exists during dental development (and thus during pulp regeneration process)—transcription factors. Notably, Msx1 is expressed in polarized preodontoblasts, whereas Msx2 is present in mature odontoblasts.[31] Protein and transcripts for Msx1 have been identified in the pulp mesenchyme at early stages of tooth development and their concentrations decreased at the bell stage.[63] The expression of these transcription factors is under the control of growth factors, and they can ultimately have broad-ranging effects. Significantly, BMP-4 upregulates Msx1 and Msx2 expression. In turn, transcription factors regulate further growth factor expression; for example, Msx1 upregulates BMP-4 synthesis in the mesenchyme and Msx2 regulates Runx2 and osteocalcin gene expression during odontogenesis.[37,38]

Growth factors and transcription factors are central to the cascade of molecular and cellular events during tooth development and are responsible for many of the temporospatial morphologic changes observed in the developing tooth germ. For these reasons, they are also likely involved in the regeneration process.

It is also important to consider the nature of the signaling process between the injurious agent and the pulp cells. Bacteria and their toxins are key candidates in the direct stimulation of pulp cells.[82] Lipopolysaccharides (LPS) and other bacterial

toxins initiate intrapulpal inflammatory processes by activation of Toll-like receptors (e.g., TLR4 activation by LPS).[76,280] Importantly, both progenitor and dental stem cells have been shown to express these receptors.[40,41,133,134,291,330] Thus stem cells within the dental pulp or periradicular tissues are equipped to detect microorganisms. Exposure of these cells to microbial antigens has been shown to directly modulate the proliferation and differentiation potential of these cells.[134,135,190] A study demonstrated that SCAPs differentiation into odontoblast-like cells was blocked when these cells were in contact with a previously infected dentin, resulting in osteoblastic differentiation instead.[308] This study showed that microbial factors can alter the ultimate differentiation fate of stem cells, reinforcing the notion that the microenvironment in which these cells are found dictates their final phenotype. Lastly, cytokines commonly found in the inflammatory milieu (including that of the dental pulp) have a profound effect on stem cells. For example, it has been recently shown that tumor necrosis factor-α (TNF-α) stimulates differentiation of dental pulp cells toward an odontoblastic phenotype via mitogen-activated protein (MAP) kinase pathway activation and p38 phosphorylation.[233,267] Therefore stem cell fate within the dental pulp is ultimately dictated by a complex cascade of intracellular signaling pathways activated by agents released from microorganisms, dentin, and immune cells.

Interestingly, morphogens are not the only naturally occurring factors found in within teeth. Several growth factors have also been evaluated for their ability to trigger the differentiation of selected MSC populations into odontoblast-like cells (Table 12.2). Interestingly, several case studies have reported that patients taking long-term corticosteroids often present with dramatic reduction of the radiographic size of the pulp chamber and up to a fivefold increase in the thickness of the predentin layer.[216,217,283] Although these patients were medically complex patients (e.g., renal failure) taking multiple drugs, the use of corticosteroids appeared to be associated with the observed increased activity of human odontoblasts. Furthermore, these unexpected "side-effects" were also observed in a retrospective study evaluating the association of pulp calcifications and the long-term use of statins.[236] These incidental effects of commonly prescribed medications were further evaluated in translational studies that have extended this general observation by demonstrating that the application of dexamethasone or statins greatly increased the differentiation of human dental pulp cells into odontoblast-like cells.[139,224] This was particularly evident when dexamethasone was combined with 1,25-dihydroxyvitamin D_3.[139] Merely changing the composition of growth factors completely altered the differentiation of these cells, with the same population of cells able to express markers of odontoblasts, chondrocytes, or adipocytes, depending on their exposure to different combinations of growth factors.[309] Such findings emphasize the importance of growth factors in guiding the differentiation of these cells. Other studies have evaluated growth factors administered alone or in various combinations for promoting differentiation of odontoblast-like cells.

Several of the approaches using compounds later found to have growth factor–like effects have immediate clinical implications. First, it is unlikely that a single growth factor will result in maximal differentiation, so combinations of growth factors may be required for evaluation in clinical

Table 12.2 Effects of Selected Growth Factors on the Differentiation of Odontoblast-Like Cells

Growth Factors	Cell Source	Phenotype	Condition	Authors
Dexamethasone	Human dental pulp	Odontoblast-like	In vitro × 8 weeks	Huang et al. (2006)[139]
Dexamethasone and vitamin D_3	Human dental pulp	Odontoblast-like	In vitro × 8 weeks	Huang et al. (2006)[139]
Dexamethasone and ascorbate-2-phosphate and β-glycerophosphate	Human or rat dental pulp	Odontoblast-like	In vitro × 3 weeks	Wei et al. (2007)[309] Zhang et al. (2005)
Insulin and Indomethacin and 3-Isobatyl-1-methylxanthine (IMBX)	Human dental pulp	Adipocyte	In vitro × 19 days	Wei et al. (2007)[309]
Dexamethasone and Insulin and Ascorbate-2-phosphate and Sodium pyruvate and TGF-β1	Human dental pulp	Chondrocyte	In vitro × 8 weeks	Wei et al. (2007)[309]
Growth/differentiation factor 11 (Gdf11)	Dental pulp	Odontoblast-like	In vitro/in vivo 10 days	Nakashima et al. (2004)
Simvastatin (statins)	Human dental pulp	Odontoblast-like	In vitro/in vivo	Okamoto et al. (2009)[224]
LIM mineralization protein 1 (LMP-1)	Human dental pulp	Odontoblast-like	In vitro/in vivo	Wang et al. (2007)
Bone morphogenetic proteins	Dental pulp	Odontoblast-like	In vitro	Saito et al. (2004) Sloan et al. (2000) Chen et al. (2008)
TGF-β1–3	Rat/monkey dental pulp	Odontoblast-like	In vitro	Sloan et al. (1999)
Demineralized dentin	Human or rodent pulp	Odontoblast-like	In vitro/in vivo	Smith et al. (1990)[274] Smith et al. (2001) Tziafas (2004)
Nerve growth factor (NGF)	Immortalized apical papilla	Odontoblast-like	In vitro	Arany et al. (2009)
Fibroblast growth factor 2	Human dental pulp	Odontoblast-like	In vitro	He et al. (2008)
Dentin matrix protein 1	Rat dental pulp	Odontoblast-like	In vivo	Almushayt et al. (2006)

trials. Related to this point, many of the studied growth factors (e.g., dexamethasone, insulin) are drugs already approved for human use in other medical/dental applications. Second, the demonstration that statins promote the differentiation of an odontoblast-like phenotype suggests that patients clinically taking statins may also have narrowing of the pulp chamber space, similar to the findings previously described for corticosteroids. This would be an important future area of research. Third, clinicians have long used demineralized human bone to augment healing after surgical procedures.[240] Demineralized human bone is thought to contain a natural combination of appropriate growth factors and scaffolds, thereby providing an appropriate environment for osteoblast differentiation or function. Extending this concept, several research groups have demonstrated that demineralized human dentin has significant benefit for promoting the differentiation of odontoblast-like cells. Importantly, translational studies in regenerative endodontics have demonstrated that irrigation of dentin with 17% EDTA increases the survival of stem cells[194,300] and odontoblastic differentiation,[106,194] possibly due to the release of bioactive molecules from dentin.[270] Collectively these findings suggest that EDTA irrigation of the dentinal walls as part of a REP could result in improved clinical outcomes.

SCAFFOLDS

An important component of tissue engineering is a physical scaffold.[176,214] Tissues are organized as three-dimensional structures, and appropriate scaffolding is necessary to (1) provide a spatially correct position of cell location and (2) regulate differentiation, proliferation, or metabolism while promoting nutrient and gaseous exchanges. It is known that extracellular matrix molecules control the differentiation of stem cells,[238,239,321] and an appropriate scaffold might selectively bind and localize cells, contain growth factors,[107] and undergo biodegradation over time.[327] Thus a scaffold is far more than a simple lattice to contain cells but instead can be viewed as the "blueprint" of the engineered tissue.

Scaffolds can be classified as either natural or synthetic. Examples of natural scaffolds include collagen,[211,213] glycosaminoglycans, hyaluronic acid (HA), demineralized or native dentin matrix,[28,57,127,212] and fibrin.[107] On the other hand, examples of synthetic scaffolds include poly-L-lactic acid (PLLA),[74] polyglycolic acid (PGA), polylactic-coglycolic acid (PLGA),[87] polyepsilon caprolactone,[322] hydroxyapatite/tricalcium phosphate,[16] bioceramics, and hydrogels such as self-assembly peptide hydrogels.[109] The great majority of currently published REPs involve evoked bleeding and the formation of a blood clot to serve as a scaffold.[77,173] Although it relatively straightforward because it does not require ex vivo manipulation, this simplistic approach is not without challenges. The blood clot is often difficult to achieve, and it does not have many of the properties of the ideal scaffold. These properties include easy delivery, adequate mechanical properties, controllable biodegradation, and incorporation of growth factors.[107] In addition, the blood clot contains a great number of hematopoietic cells, which eventually undergo cell death releasing their toxic intracellular enzymes in the microenvironment that may be detrimental to stem cell survival.

Another approach for creating a scaffold involves the use of autologous PRP. It requires minimal ex vivo manipulation, being fairly easy to prepare in a dental setting. PRP is rich in growth factors, degrades over time, and forms a three-dimensional fibrin matrix.[20,148,223] Platelet-rich fibrin (PRF) is an alternative to PRP because it has a three-dimensional architecture conducive with stem cell proliferation and differentiation and contains bioactive molecules.[73,75] These autologous scaffolds have been used successfully in regenerative cases.[173] However, it should be emphasized that, despite their reported use, there are some drawbacks to their clinical use; they require collection of intravenous blood that can be challenging in children, the diversity and concentration of growth factors within PRP and PRF preparations are not controllable, and they lack temporal degradation control and the mechanical strength to support a coronal restoration. Thus, despite some desirable characteristics, other scaffold alternatives to PRP and PRF should be carefully considered.

Hydrogels are a class of scaffolds composed of three-dimensional hydrophilic polymers that absorb water or tissue fluids up to several times their weight.[94,109] These water-swollen materials are easily injectable in their colloidal form, undergoing gelation by chemical (e.g., changes in pH and osmolarity) and/or physical (e.g., temperature change) cues. These materials are highly tunable and biocompatible and can be designed to resemble naturally occurring extracellular matrices.[319] They are of particular interest for regenerative endodontics because they can be easily injected into narrow root canal spaces and be modified to deliver chemotactic and angiogenic agents to drive stem cell homing and supportive angiogenesis.[77,121] Hydrogels made of self-assembly peptides (e.g., Puramatrix)[54] show great potential to be used in endodontic tissue engineering because their sequence includes short peptide sequences, similar to those naturally occurring in tissues, enhancing cell attachment and proliferation.[111] Similarly, fibrin-based scaffolds have been shown to have excellent properties that they may be well suited for use in regenerative endodontics. These scaffolds have the potential to be functionalized because they can be loaded with growth factors or dentin matrix protein extracts that promote in vitro and in vivo proliferation and proper differentiation of stem cells.[103,311] In addition, fibrin is a natural scaffold that already is used in medicine (e.g., surgical fibrin glue) and could potentially be adopted for use in REPs.

Decellularized human and animal tissues have been used as scaffolds for many decades in regenerative medicine and dentistry.[116] An adequate decellularization procedure is needed to obtain a suitable ECM scaffold from native tissues. It should minimize the loss or damage of the extracellular matrix (ECM) components and maximize the removal of the cellular component to reduce the immunologic responses.[116,316] There are many examples of decellularized tissues that have been approved by the US Food and Drug Administration (FDA) to be used as scaffolds, including dermal allograft (Alloderm), human peripheral nerves (Avance Nerve Graft, Axogen), porcine small intestine submucosa (SurgiSIS, Cook Biotech), porcine urinary bladder (MatriStem, Acell), and porcine heart valves (Syntergraft, Cryolife).[12,44] Unfortunately, these scaffolds do not share the intricate ultrastructure and molecular composition of the dental pulp. A study demonstrated that human dental

pulp from extracted healthy third molars can be successfully decellularized and used as a scaffold.[277] SCAPs, when seeded in the decellularized dental pulp scaffold, proliferated in three dimensions and differentiated into odontoblast-like cells extending cellular processes into dentinal tubules (Fig. 12.7).[277] This study introduced the concept that harvested dental pulp, from otherwise discarded extracted molars, could be made into scaffolds for use in regenerative endodontics, similarly to many other tissues currently available for tissue specific regeneration in medicine. The results of this study have been confirmed by another study that evaluated the use of decellularized swine dental pulp in REPs.[14] The authors demonstrated that the decellularized scaffold allowed for cell proliferation in vitro and that its implantation into pulpectomized teeth in dogs favored the recruitment of apical stem cells resulting in the formation of a vascularized pulplike tissue with expression of an odontoblastic marker (i.e., DSP).[14] Collectively, these studies demonstrated that, similarly to other tissues, dental pulp could be decellularized and used as scaffold for regenerative purposes. This allograft approach is further facilitated by the readily available source of this tissue because millions of healthy third molars are extracted and discarded every year in the United States alone. Alternatively, dental pulp from animals such as swine could be used as the source of a xenograft scaffold.

DELIVERY SYSTEM

Even with selection of the appropriate cell source, growth factors, and scaffold, the resultant mixture must be delivered in a spatially appropriate fashion into the space of the root canal system. For example, nearly all cells of the body are within 0.1 to 1 mm of a blood vessel to maintain adequate diffusion of oxygen and nutrients.[122,136] This represents a challenge still to be overcome in the currently performed REPs that recruit stem cells[58,191] to a canal space devoid of lateral vascularity and several millimeters away from apical blood vessels. If one were to inject cells, in a cell-based approach, along the entire coronal-apical extent of a root canal system, the vast majority of cells would be expected to succumb to tissue hypoxia. Interestingly, it has been demonstrated that, under hypoxic conditions, stem cells proliferate faster and release greater levels of angiogenic factors such as vascular endothelial growth factor 1 (VEGF-1) that promote targeted angiogenesis into the engineered space.[21] Thus an alternative approach would be to inject a scaffold with chemotactic factors into the root canal. This approach is called cell homing because cells are attracted to the scaffold along with supportive blood vessels in a progressive manner;[170] instead of being abruptly delivered to an avascular space (i.e., similar to the current revascularization procedures), the cell-homing approach can be applied in a cell-free[108] (no cells

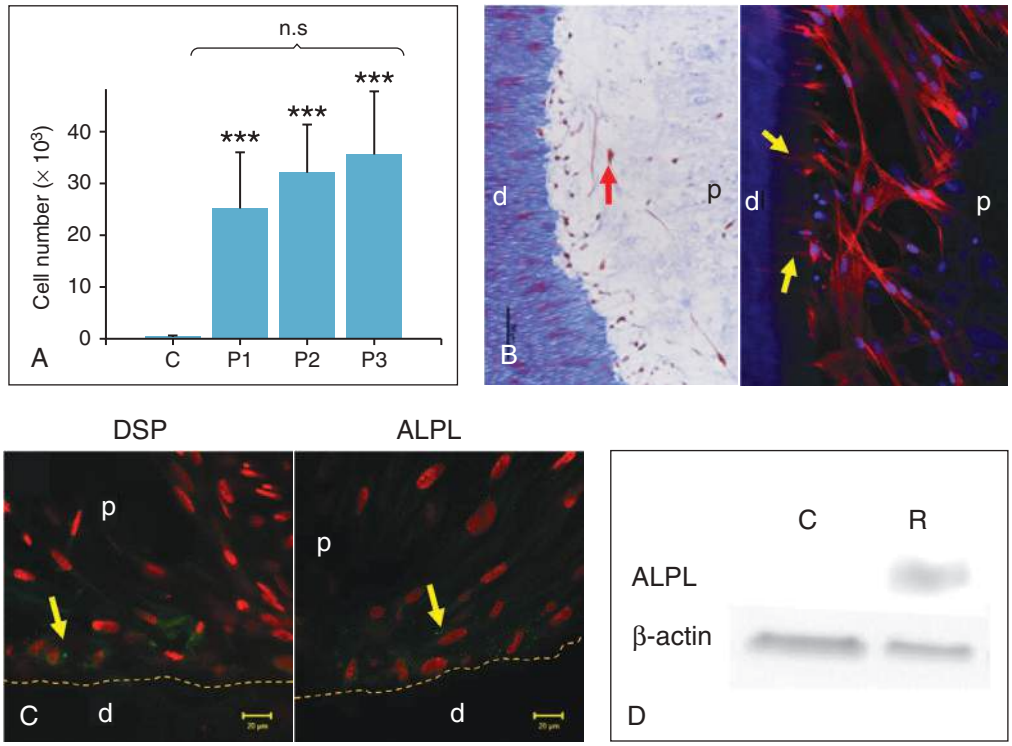


Fig. 12.7 SCAP repopulation in decellularized dental pulp scaffold. (**A**) Cell viability in repopularized human dental pulp (rHDP) at 2 weeks after cell seeding. The data are presented as mean number of viable cells and standard deviation values. ***$P < .001$ with one-way ANOVA analysis followed by post hoc Bonferroni test compared with control (b) Masson trichrome staining (a) and fluorescent staining with 4′,6-diamidino-2-phenylindole (DAPI) and phalloidin at 2 weeks after cell seeding. *Arrowhead* in left side indicates repopulated cells which have spindle-like shape, those in the right side indicate odontoblastic processes (**B**). There was no statistically significance between other experimental groups. (**C**) Immunofluorescence cell staining at 8 weeks after cell seeding for dentin sialoprotein (DSP) and alkaline phosphatase, liver/bone/kidney (ALPL). Positive signals *(green)* were mainly found in the cells adjacent to dentin. (**D**) Western blot analysis at 8 weeks after cell seeding revealed elevation of ALPL expression in rHDP group. Scale bars: 50 μm in **B** and 20 μm in **C** and **D.** *C,* Control; *P1,* protocol 1; *P2,* protocol 2; *P3,* protocol 3; *p,* dental pulp; *d,* dentin; *R,* repopulation. (Reproduced with permission from Song JS, Takimoto K, Jeon M, et al: Decellularized human dental pulp as a scaffold for regenerative endodontics, *J Dent Res* 96[6]:640–646, 2017.)

implanted along the chemotactic factors) (Fig. 12.8) or cell-based approach (cells are delivered in the chemotactic-containing scaffold).[77,208] Because dental pulp can be approximated as a loose connective tissue core surrounded by a layer of odontoblasts, the spatial arrangement of cells and growth factors within the scaffold may be particularly important to promote odontogenesis without having complete calcification of the root canal system. Complete recapitulation of the pulp-dentin complex architecture requires additional research effort.

TRANSLATIONAL STUDIES

Several elegant studies in regenerative endodontics have used various translational methodologies, including evaluation of clinical samples,[59,191] organotype root canal models,[194,300] tooth slice models,[74,120,251] whole tooth culture,[178,286] and animal models.[106,170,212,213,295,296] These studies have been crucial to provide a strong scientific foundation for the field of regenerative endodontics while allowing for clinical treatment protocol optimization, and the development of new treatment strategies such as inclusion of scaffold and growth factors in regenerative procedures.[211,219]

A study demonstrated that new dentin and pulplike tissue could be generated in human root segments implanted subcutaneously in immunocompromised mice.[141] In this study, root segments had one of the openings sealed with MTA to mimic the coronal restoration of regenerative endodontic cases. The canal space was filled with either SCAPs or DPSCs in a PLGA-based scaffold. The implants were harvested 3 months later and processed for immunohistochemical analysis. The results indicated that there was a dramatic circumferential apposition of dentinlike material along the dentinal walls. The new mineralized tissue was lined with polarized cells expressing odontoblastic markers. In addition, the dentinlike tissue was largely atubular and displayed cellular inclusions similar to the histologic presentation of osteodentin. Importantly, the cells from the engineered pulp were positive for human mitochondria demonstrating that they were originated from the implanted human stem cells and not from the host (mouse). Lastly, the root segments that were implanted without stem cells had only connective tissue that did not resemble a pulplike tissue nor had mineralized tissues and odontoblast-like cells. Thus a pulp-dentin complex could be engineered in human roots implanted subcutaneously in immunodeficient mice.

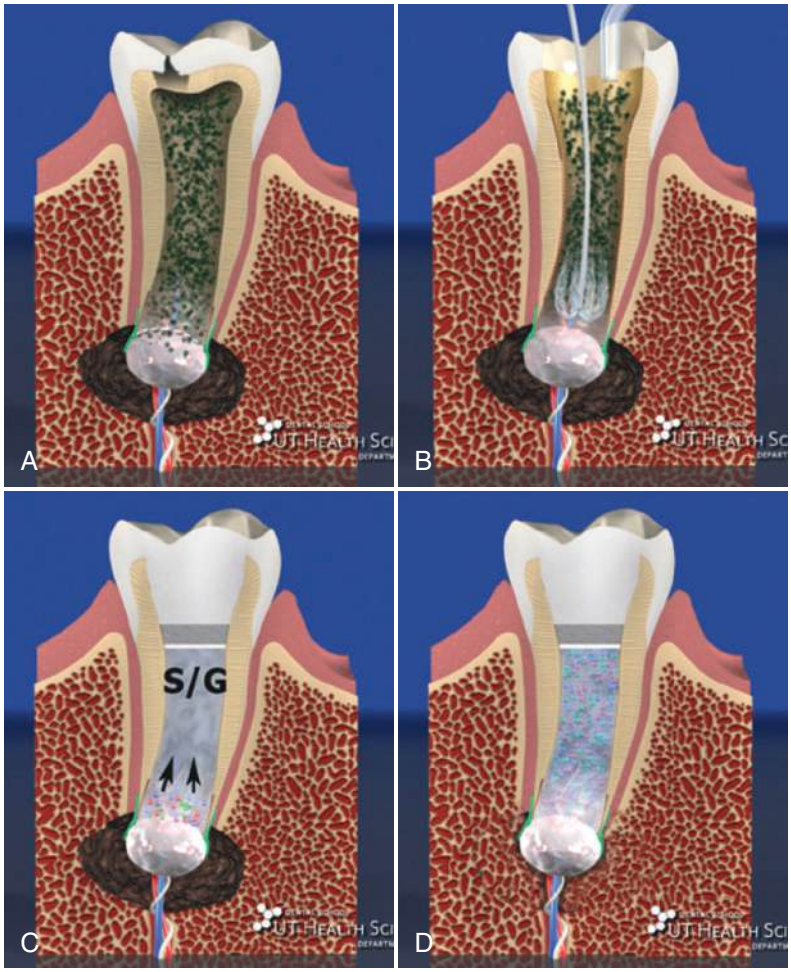


Fig. 12.8 Schematic drawing illustrating bioengineering using a cell homing approach. An immature premolar with a necrotic pulp and apical lesion **(A)** is disinfected **(B),** followed by placement of a biodegradable scaffold *(S)* containing growth factors and chemotactic factors *(G)* to allow progressive proliferation and migration of apical stem cells into the canal space **(C)** leading to the population of the canal space with stem cells concomitantly with vascular supply and tissue organization **(D).** (From Diogenes A, Henry MA, Teixeira FB, Hargreaves KM: An update on clinical regenerative endodontics, *Endod Topics* 28:2–23, 2013.)

Complete pulpal regeneration in dogs has been accomplished by Nakashima and colleagues.[142] In this elegant study, dental pulp was removed via sterile pulpectomy procedure, followed by placement of sorted CD105+ DPSCs in a collagen gel to the midroot. The remaining coronal part of the canal was back-filled with the collagen gel containing the chemotactic factor stromal-derived factor 1 (SDF-1). Subsequent histology demonstrated formation of new pulp tissue with innervation, vascularization, and odontoblast-like cells lining the dentinal walls. In addition, the engineered pulp had protein and RNA expression similar to the native dental pulp.[142]

In another important study, mobilized DPSCs were generated by selecting DPSCs that migrated towards a concentration gradient of granulocyte colony-stimulating factor (G-CSF). These selected cells were implanted in a collagen gel into pulpectomized root canals.[143] Complete pulpal regeneration was observed, with evidence of new dentin formation, blood vessels, and innervation in the engineered tissue (Fig. 12.9). The impressive results with this cell-based approach have laid the foundation for the use of this technology in emerging clinical trials using stem cell transplantation.[215]

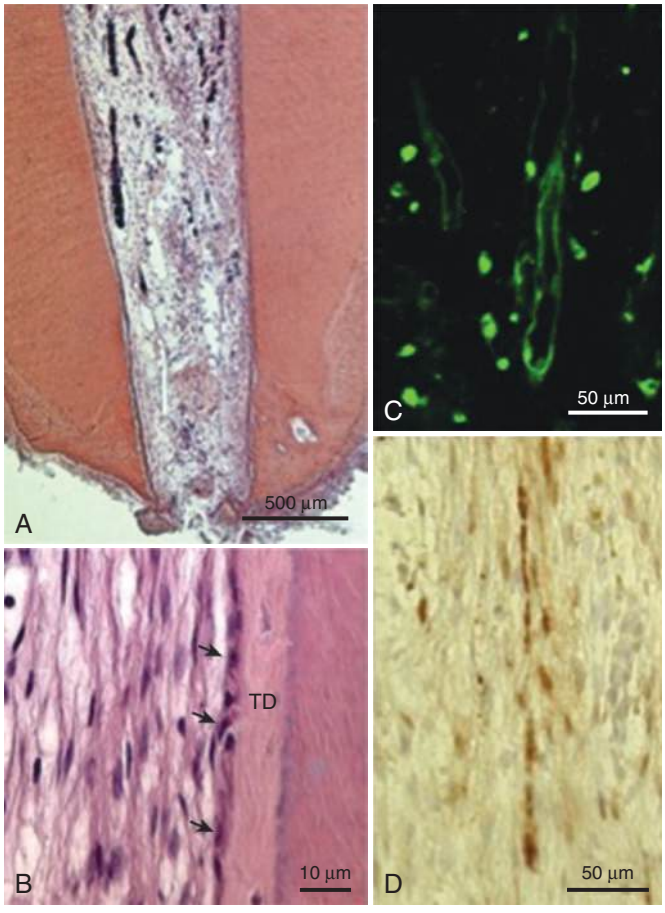


Fig. 12.9 Regeneration of pulp tissue after autologous transplantation of mobilized dental pulp stem cells (MDPSCs) with G-CSF in pulpectomized teeth of young dogs on day 14. (**A** and **B**) Regenerated pulp tissue. (**B**) Odontoblastic cells *(black arrows)* lining to newly formed osteodentin/tubular dentin *(TD)* along with the dentin. (**C**) Immunohistochemical staining of BS-1 lectin. (**D**) Immunohistochemical staining of PGP9.5. (From Nakashima M, Iohara K: Mobilized dental pulp stem cells for pulp regeneration, *J Endod* 40:S29, 2014.)

It is noteworthy that all the demonstrations of pulp regeneration in animal models were in root canals without history of infection and pulpal necrosis.[142-144,251] Previously infected root canals must be adequately disinfected to suppress chronic inflammation that is detrimental to regeneration.[62] However, many irrigants and medicaments have detrimental effects on stem cell survival and differentiation.[79] Several studies using organotype root canal models and animal models evaluated the combination and concentrations of irrigants and medicaments that allowed for stem cells proliferation and differentiation.[15,177,194,300,310,312] Furthermore, emerging studies demonstrated that bacterial antigens have a robust effect on the terminal differentiation of stem cells guiding these to a greater osteogenic as opposed to an odontoblastic phenotype.[40,308] Thus adequate disinfection, resolution of inflammation, and the detoxification of residual microbial antigens appear to be limiting factors in complete pulpal regeneration, despite exciting results from current advances in dental pulp tissue engineering.[79]

SUMMARY OF BASIC RESEARCH ON REGENERATIVE ENDODONTICS

Regeneration of a functional pulp-dentin complex relies on the foundation of tissue engineering and can be viewed as a function of the spatially correct delivery of appropriate stem cells and growth factors embedded within a scaffold. Although considerable research has used in vitro cell culture methods to identify key factors regulating the differentiation of odontoblast-like cells, it should be noted that emerging studies conducted in animal models are quite promising for regeneration of this pulp tissue. Preclinical studies involving surgical placement of a human tooth filled with human stem cell/growth factor/scaffold combination into immunocompromised mice[74,106,251] have permitted histologic analysis of neovascularization as well as the differentiation and mineralization activity of newly formed odontoblasts. The use of human cells in a mouse model permits histologic confirmation that the resulting odontoblast-like cells were of human origin. These novel findings provide strong evidence that either human SCAP or DPSC cell sources, on a PLGA scaffold, were able to regenerate a vascularized tissue that had histologic evidence of odontoblast-like cell differentiation and the spatially appropriate formation of dentinlike material onto the root canal walls. Although no specific growth factors were added to this mixture, it is important to note that the root canal walls were treated with 17% EDTA, an irrigant known to expose endogenous growth factor proteins embedded in the dentinal walls.[332] This and other related studies provide strong impetus for clinical translational research evaluating various potential regenerative endodontic therapies.

Clinical Studies on Regenerative Endodontics

To date, most case reports, case series, and retrospective studies published in regenerative endodontics have not fully incorporated the tissue-engineering concepts described. Instead, most of these reports present cases with variations of revascularization techniques.[77] These procedures were

initially performed empirically with a strong focus on disinfection and the intentional bleeding into the root canal. However, it became obvious that these procedures were in fact stem cell–based procedures with all the three components of the tissue-engineering triad present: stem cells,[58,191] growth factors,[32] and scaffolds.[27,173,293] Important preclinical studies previously described in this chapter have provided the foundational framework for a paradigm shift. This shift represents a clear departure from the traditional "disinfect the canals at all cost" to "disinfection while creating a microenvironment conducive for tissue engineering." Varied terminologies have been given to these procedures that include revascularization,[27] revitalization,[155,181,186,195,299] and maturogenesis.[5,115,210] Of these terms, the most popularly used has been revascularization. This term is largely based on the trauma literature observation that immature teeth could become "revascularized" after trauma. However, revascularization is a term better used for the reestablishment of the vascularity of an ischemic tissue, such as the dental pulp of an avulsed tooth. From this perspective, a focus on revascularization would ignore the potential importance of growth factors and scaffolds that are required for histologic recapitulation of the pulp-dentin complex. Although we appreciate that angiogenesis and the establishment of a functional blood supply are a key requirement in the maintenance and maturation of a regenerating tissue, it is noteworthy that some of the published cases report positive responses to pulp sensitivity tests such as cold or electric pulp test.[77] In addition, histologic analysis of human teeth treated with these procedures revealed the rich presence of innervation that includes nociceptors.[24,179] This is evidence that a space that was previously vacant (débrided root canal) may become populated with an innervated tissue supported by vascularity. Furthermore, studies that performed histologic analyses of the tissue formed following these procedures in patients revealed the presence of ectopic bone formation, periodontal ligament, and cementum in addition to vascularized loose connective tissue.[30] However, odontoblast-like cells and newly formed tissue resembling dentin have also been found in certain cases.[24,234,263] Taken together, the core concepts of tissue engineering distinguish a regenerative treatment philosophy from a revascularization philosophy derived from certain trauma cases (which only occur in a low percentage of replanted teeth). Lastly, there have been a number of reports of procedures performed with intentional manipulation of the principles of tissue engineering such as use of autologous PRP,[9,112,151,237,293] PRF,[206,210] and exogenous growth factors and scaffolds.[173] Thus, instead of using different terms for each variant of these procedures, we will refer to them simply as REPs that include past, present, and future procedures that aim to the functional regeneration of the pulp-dentin complex.

CLINICAL PROCEDURES RELATED TO REGENERATIVE ENDODONTICS

There are several challenges clinicians face when presented with an incompletely formed root in need of endodontic treatment.[77] Because the apex is not fully developed and often has a blunderbuss shape, cleaning and shaping of the apical portion of the root canal system can be difficult. The process is further complicated by the presence of thin, fragile dentinal walls that may be prone to fracture during instrumentation or obturation. In addition, the open apex increases the risk of extruding material into the periradicular tissues. Traditionally an immature tooth with an open apex is treated by apexification, which involves creating an apical barrier to prevent extrusion. In many cases, this entails an involved, long-term treatment with $Ca(OH)_2$, resulting in the formation of a hard-tissue apical barrier.[65,66,99] However, a disadvantage of the traditional apexification procedures is that the short-term or long-term use of $Ca(OH)_2$ has the potential to reduce root strength.[29,80,163,250,269,304,323] This finding is consistent with a large case series using the traditional apexification protocol; it showed that a major reason for tooth loss following apexification was root fracture.[65,152] In a retrospective study, the use of calcium hydroxide in apexification procedures resulted in the fracture of 23% of the teeth treated during the follow-up period of up to 18 months.[152] The advent of one-step apexification, by creation of artificial barriers (i.e., apical plugs) using materials such as MTA,[196,197,314,315] has greatly decreased the number of appointments and time to completion. Importantly, the one-step apexification has been shown to have as high success rate as apexification with calcium hydroxide in resolving apical periodontitis (both symptoms and radiographic presentation).[297] However, apexification procedures do not generally result in further root development.[39] A primary advantage of REPs in these cases is the greater likelihood there will be an increase in root length and root wall thickness, in addition to the possible regain of vitality responses. In addition to managing the open apex, REPs have been shown to be beneficial in cases of root resorption promoting the arrestment of the resorptive process and repair of the resorbed root (Fig. 12.10).[164,254,301]

There have been numerous published cases of REPs. Investigators and clinicians have used a variety of medicaments to disinfect the canal space.[77] Between 51% and 80% of the cases included the use of a triple antibiotic paste (TAP) (a 1:1:1 mixture of ciprofloxacin/metronidazole/minocycline), whereas 20% to 37% used $Ca(OH)_2$ as an intracanal medicament.[77,173]

The development of the TAP was led in large part by Hoshino and colleagues.[258,259] They demonstrated the effectiveness of combinations of antibiotics (and in particular the high efficacy of the combination of ciprofloxacin, metronidazole, and minocycline) in eradicating bacteria from the infected dentin of root canals.[258] Astute practitioners realized that the TAP could be a valuable adjunct for revascularization procedures because it could be used to create an environment favorable for the ingrowth of vasculature and regenerative cells by reducing or eradicating bacteria in the canal space of teeth with necrotic pulps and incompletely formed apices. The efficacy of the TAP in disinfecting necrotic root canal systems has been demonstrated in a preclinical model.[313] In this dog study, 60 teeth were accessed and infected by sealing dental plaque and sterile saline on a cotton pellet into the pulp chamber for 6 weeks. By the end of this period, each premolar was radiographically confirmed to have apical periodontitis. The canals were then sampled at three time points: before and after irrigation with 1.25% NaOCl, and 2 weeks after the delivery of the TAP into the root canal system using a Lentulo spiral.

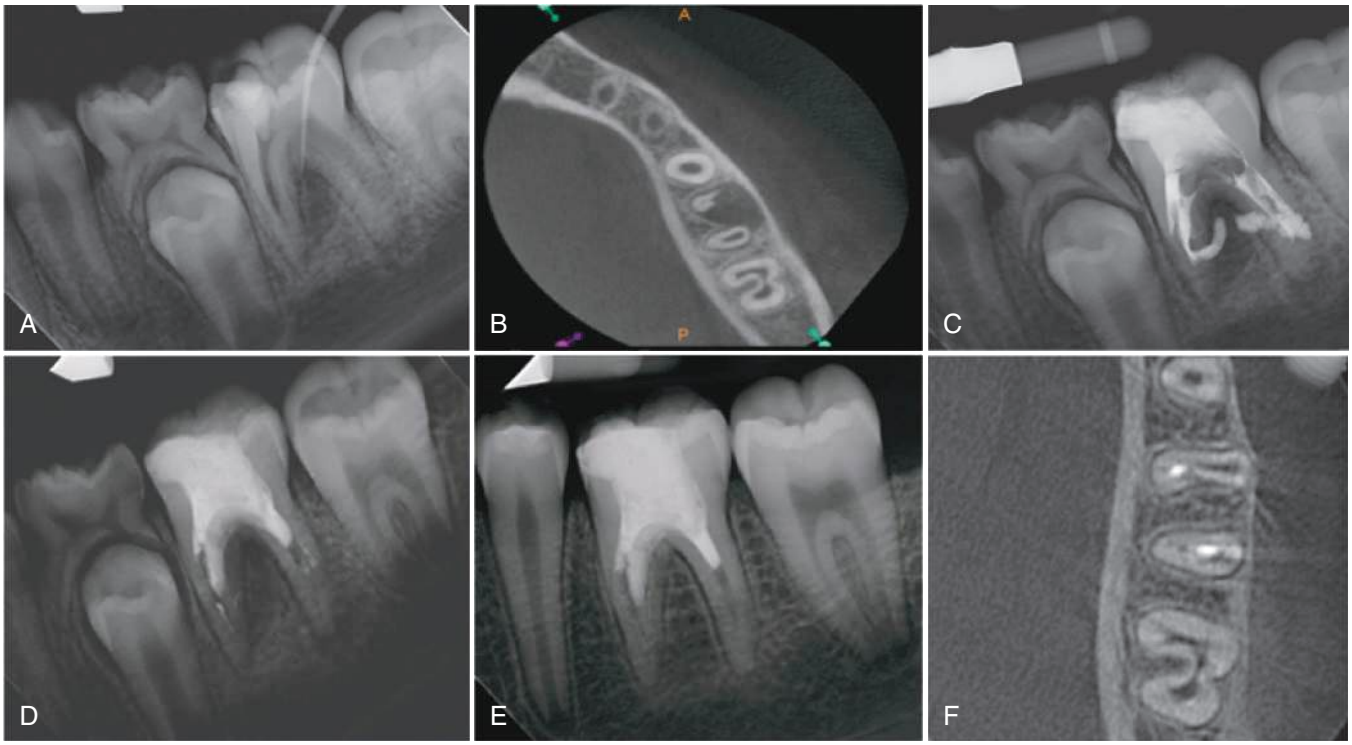


Fig. 12.10 An 11-year-old female was referred for evaluation of tooth #19 diagnosed with pulp necrosis and chronic apical abscess traced with a gutta-percha cone as seen in the periapical (PA) radiograph (**A**). A small-volume CBCT revealed resorption in mesial and distal roots (**B**). The canals were disinfected and medicated with Vitapex (**C**). Four weeks later, the patient was asymptomatic and tooth #19 and the sinus tract healed. The tooth was reaccessed, irrigated with 1.5% NaOCl and 17% EDTA, and bleeding evoked in all canals precurved 25 K. MTA was placed over the blood clot followed by glass ionomer and composite restoration (**D**). After 24 months, the patient was asymptomatic, the tooth responded to electric pulp tester (EPT) demonstrated complete healing as seen in PA radiograph (**E**) and small-volume CBCT (**F**). (Case courtesy of Dr. Ahmed Alelyani, University of Texas Health Science Center.)

Before irrigation, all of the teeth had positive cultures for anaerobic bacteria, with a mean colony-forming unit (CFU) count of 1.7×10^8. After irrigation with 1.25% NaOCl, 10% of the teeth sampled cultured bacteria free. The mean CFU count was 1.4×10^4, or an approximate 10,000-fold reduction in viable bacteria. After dressing with the TAP for 2 weeks, 70% of the teeth sampled cultured bacteria free. The mean CFU count was only 26, which is approximately another 1000-fold drop in bacteria. These findings were confirmed in another related dog study.[289] This study provides strong support for the effectiveness of the TAP in disinfection of immature teeth with apical periodontitis.

As stated previously, calcium hydroxide has been the second most used intracanal medicament in published cases. This application represents a new use of a long established intracanal medicament in endodontics. Although $Ca(OH)_2$ appears to be less effective against some intracanal bacterial species than antibiotic paste formulations,[22,249] its use is associated with lower cytotoxicity to stem cells,[15,247] release of important bioactive growth factors from the treated dentin,[128] and greater survival and proliferation of stem cells in presence of the conditioned dentin.[113] In addition, the relatively short-term use of this medicament in regenerative procedures does not appear sufficient to reduce fracture resistance.[323] Another factor to consider when choosing an intracanal medicament is the ability to remove the medicament from the canal space. One study that addressed this question incorporated radioactive tracers in both calcium hydroxide paste (Ultracal; Ultradent, Inc.) and TAP (Champs Pharmacy; San Antonio, TX).[36] The radiolabeled medicaments were placed in extracted teeth with standardized root canals. After 28 days of incubation, canal spaces were irrigated with a standardized protocol using different techniques. Surprisingly, greater than 80% of the TAP could not be removed from the tooth (Fig. 12.11), and it was found not in the canal lumen but greater than 350 μm into the dentinal tubules. In contrast, greater than 80% of calcium hydroxide was removed (see Fig. 12.11) with the remaining medicament present in superficially within dentin.[36] This is an important finding, given that drugs remaining within dentin are likely to have an effect on the fate of stem cells in contact with the treated dentin. Another study revealed that TAP, when used at concentrations typically used in case reports (1 g/mL), and removed with standardized irrigation protocol, resulted in no SCAP survival.[15] In contrast, dentin treated with calcium hydroxide promoted SCAP survival and proliferation, and dentin treated with TAP at a 1 mg/mL had no effect on SCAP survival (Fig. 12.12). The ultimate success of REPs relies on maximizing disinfection and detoxification of the root canal system while minimizing detrimental effect on survival and differentiation of stem cells. Therefore it is imperative that the choice of medicament and concentration must also be antimicrobial. The concentration of 1 mg/mL of TAP, approximately 1000 times less concentrated than commonly used in the form as paste, has been tested for its antimicrobial efficacy in a randomized clinical trial using qRT-PCR to detect residual bacteria.[22] In that study, this concentration

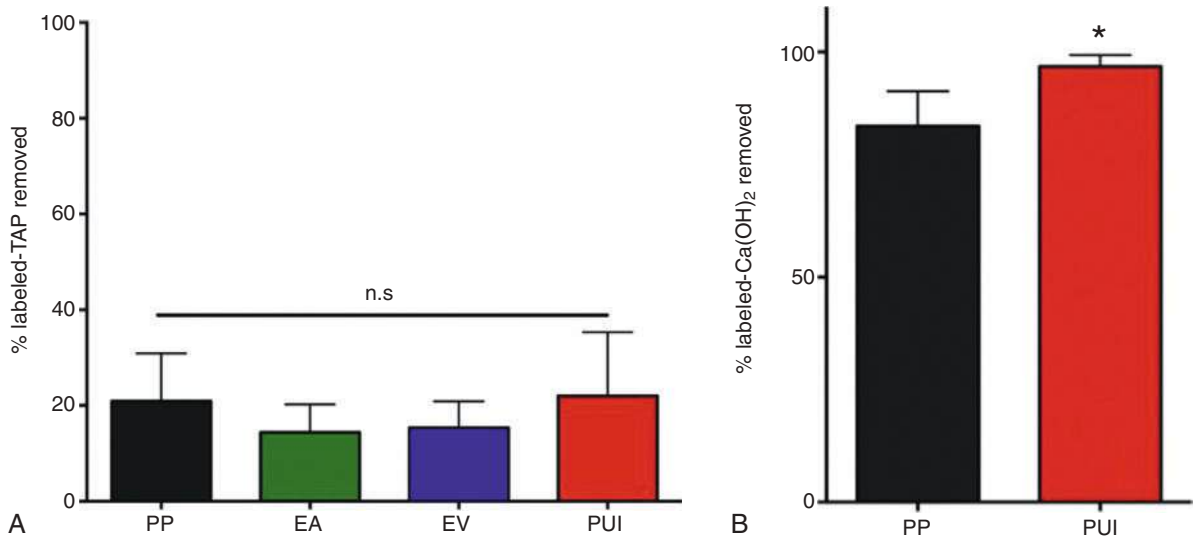


Fig. 12.11 TAP remains in dentin, whereas most $Ca(OH)_2$ is eliminated after endodontic irrigation. Radiolabeled TAP or $Ca(OH)_2$ was placed within canals of standardized root segments and incubated for 28 days at 37°C. The canals were flushed with standardized volumes of EDTA and saline using either positive pressure with a side-vented needle (PP) or positive pressure with ultrasonic activation of irrigants (PUI). There was no difference in labeled TAP removal among groups with only approximately 20% of the medicament being removed by the irrigation protocols (**A**). In contrast, >80% of $Ca(OH)_2$ was removed, with more efficient removal observed in canals irrigated with PUI (**B**). Data are presented as the mean percentage of total radiolabeled medicament removal ± standard error of the mean. *$P < .05$ tested by the Student *t* test ($n = 12$/group). (Modified with permission from Berkhoff JA, Chen PB, Teixeira FB, Diogenes A: Evaluation of triple antibiotic paste removal by different irrigation procedures, *J Endod* 40:1172, 2014.)

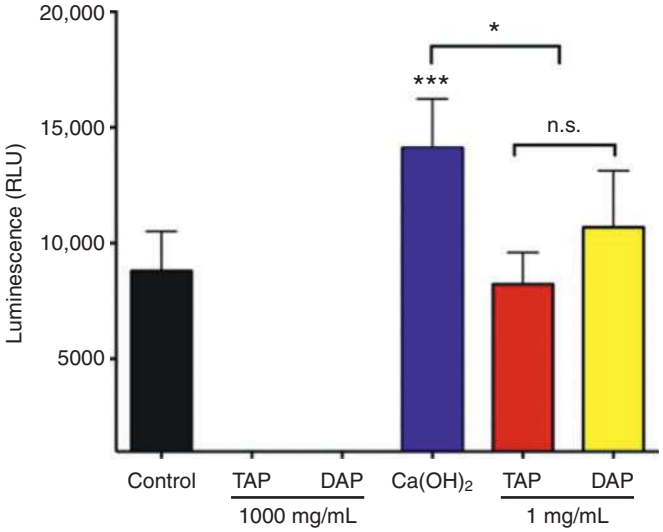


Fig. 12.12 Dentin conditioning for 7 days with medicaments used in REPs has a profound effect on SCAP survival. Standardized dentin disks were treated for 7 days with TAP or double antibiotic paste (DAP) (concentrations of 1000 mg/mL or 1 mg/mL), $Ca(OH)_2$ (Ultracal), or sterile saline (control). SCAP in a Matrigel scaffold (BD Biosciences; Bedford, MA) was seeded into the lumen of the disks after the medicaments were removed and cultured for 7 days. Cell viability (survival) was determined using a luminescent assay. SCAP culture on dentin treated with TAP or DAP at the concentration of 1000 mg/mL resulted in no viable cells. Conversely, dentin conditioning with TAP or DAP at the concentration of 1 mg/mL supported cell viability with no difference from untreated dentin disks (control). Greater survival and proliferation were detected in the group treated with $Ca(OH)_2$. Data are presented as mean ± standard deviation of relative luminescence units ($n = 12$/group). *$P < .05$; ***$P < .001$; n.s., no statistical difference as tested by 1-way analysis of variance. (From Althumairy RI, Teixeira FB, Diogenes A: Effect of dentin conditioning with intracanal medicaments on survival of stem cells of apical papilla, *J Endod* 40:521, 2014.)

of TAP resulted in significantly less measurable bacteria than a mixture of calcium hydroxide and chlorhexidine.[22] This clinical study confirmed the findings of many in vitro studies demonstrating that the effective antibacterial concentration of either triple or double antibiotic is 1 to 10 mg/mL.[150,154,177,249,284] It is important to mention that at the concentration of 1 to 10 mg/mL in aqueous medium, the antibiotic formulation is a solution. Despite this solution being effective clinically,[22] its placement and bioavailability over 2 to 4 weeks is more challenging than in a paste form. However, this concentration can be easily used if combined with a macrogol and propylene glycol pasty vehicle as first described by Hoshino and colleagues to facilitate placement and enhance dentinal tubule penetration.[64,138,259] Alternatively, the use of a hydrogel solution has been investigated and methylcellulose found to be a suitable vehicle for delivery of lower concentration of antibiotic mixtures,[284] and it can be combined with a radiopaque agent to allow for radiographic confirmation of its adequate placement.[307] Lastly, resorbable drug delivery systems are being developed to allow for localized and controlled drug release of predefined concentrations of drugs that are known to be antibacterial while less detrimental to the microenvironment and cells involved in the regenerative process.[10,42,162] Thus clinicians must carefully evaluate the advantages and disadvantages of each intracanal medicament while observing the ideal therapeutic concentration.[79]

Regardless of the intracanal medicament used, REPs have common features. Most of these published procedures reported minimal to no instrumentation.[77] This might be due, at least in part, to the concern of further weakening fragile dentinal walls and the difficulty of mechanically debriding canals of such large diameters. Due to the lack of

mechanical debridement, clinicians relied on copious irrigation for maximum antimicrobial and tissue dissolution effects.[77] Canals were then medicated for a period that varied from days to several weeks. At the second visit, if signs and symptoms of diseased had subsided, the medicament was removed, the canal was dried, and intracanal bleeding was evoked in most, but not all, cases.[77] In most cases a coronal plug of MTA was placed over the blood clot or a collagen-based internal matrix, followed by a bonded coronal restoration.[173] Each of these steps will be discussed later in light of the current American Association of Endodontists (AAE) considerations for REPs.

OVERVIEW OF CLINICAL REGENERATIVE ENDODONTIC PROCEDURES

To date, most published clinical studies on REPs are composed of case reports and case series, and fewer retrospective cohort study[13,60,152,198,265,278] and prospective randomized clinical trials[185,211] on regenerative endodontics treatment are available. Although case series do not provide definitive evidence to support a given treatment modality, they do have the advantage of being conducted in actual patients and thus provide a higher level of evidence than preclinical studies. Although techniques for regenerative endodontics have varied in published case reports and case series,[77] there have been some consistent features worth noting. Nearly all reported cases involve patients 8 to 18 years old and teeth with immature apices with exception of published cases performed in mature, fully formed teeth.[132,151,209,252–255] Thus the patient age appears to be an important factor in case selection because some studies suggest that younger patients have a greater healing capacity or stem cell regenerative potential. Another important factor related to age is the stage of root development because the large diameter of the immature (open) apex may foster the ingrowth of tissue into the root canal space and may be indicative of a rich source of mesenchymal stem cells of the apical papilla (SCAP; see Figs. 12.2 and 12.3). These tissues are likely lacerated during the evoked-bleeding step and constitute a likely source of MSCs delivered into the root canal space. Indeed, a clinical study comparing outcomes in REPs found that greater root development was detected in the younger patient group, particularly with apices measuring more than 1 mm on the periapical radiographs.[90] Another consistent finding reported in nearly every case is the lack of instrumentation of the dentinal walls related to concerns about the potential fracture of these thin, incompletely developed roots. The lack of instrumentation would be expected to have the benefit of avoiding generation of a smear layer that could occlude the dentinal walls or tubules. On the other hand, the lack of instrumentation could result in remaining bacterial biofilms within dentinal tubes. This is particularly an issue in immature teeth due to large patent dentinal tubules that are more prone to bacterial penetration.[161] This is an issue that has not been evaluated in many cases, but the lack of canal wall instrumentation (and subsequent identification of bacteria in the apical dentin) has been suggested as a reason for failure of a regeneration case.[187] It is important to emphasize that these procedures are marked by robust disinfection protocols. Sodium hypochlorite, either alone or in combination with other irrigants, has been used to disinfect the canal space in most cases.[173] In a majority of cases, a combination of triple antibiotic (minocycline, metronidazole, and ciprofloxacin) was left in the canal space for a period of days to weeks, so the disinfection protocol was primarily a chemical method rather than the chemomechanical approach used in conventional nonsurgical endodontic therapy.[77] In most cases a blood clot was formed in the canal and might serve as a protein scaffold, permitting three-dimensional ingrowth of tissue.[77,173] However, it should be noted that few published cases included the use of PRP or platelet-rich fibrin (PRF)[148] and the use of exogenous scaffolds such as collagen type 1[211] and hydroxyapatite.[219]

Nearly all of these reports noted continued thickening of the root walls and subsequent apical closure. Examples of increased root length and thickening of root walls following REPs are seen in Figs. 12.13–12.16. Because of the lack of histology in most clinical cases, it should be recognized that radiographic findings of continued root wall thickness do not necessarily indicate that dentin was formed. Based on histologic results from preclinical studies, it is possible that the radiographic appearance of increased root wall thickness might be due to ingrowth of cementum, bone, or a dentinlike material. Histologic evidence from human extracted teeth following REPs suggests that regenerated pulp tissue may be in the canal space[263] and that the mineralized tissue along the dentinal walls appear to be cementum-like or osteodentin.[195,199,256] It should also be noted in some of the case reports that, although the teeth were nonresponsive to pulp testing, the authors suggested that vital tissue was identified in the apical portion of the canal space. In these cases, necrotic tissue was removed until bleeding was observed, then the canals were disinfected with antibiotic paste or $Ca(OH)_2$.[77] One could argue that by leaving vital pulp tissue in the apical segment of the canal, the resulting progression of root formation is more similar to apexogenesis than revascularization. Although the biologic process may differ between the revascularization and apexogenesis cases presented in these case reports, the goals of the procedures are similarly advantageous and significant. In both procedures, there is healing of the periradicular tissues and a progression of root development in a tooth that would otherwise have had a progression of pulpal and periradicular pathosis.

A retrospective study has compared the radiographic changes in 48 revascularization cases to 40 control cases.[39] Although published revascularization cases have been treated with varying clinical protocols, they can be grouped by method of canal disinfection—namely, TAP, $Ca(OH)_2$ treatment, or formocresol treatment. This study applied a mathematic image-correction procedure that permitted the comparison of nonstandardized radiographs with subsequent statistical analysis of radiographic outcomes. The percent change in root dimensions was first compared in two negative control groups (nonsurgical root canal treatment [NSRCT] and MTA apexification) predicted to have little to no change in root dimensions.[39] This provides an internal test that the mathematic analysis was appropriate. The results indicate that these two negative control groups had minimal measured changes in root width (Fig. 12.17) or root length (Fig. 12.18), with the anticipated finding that instrumentation with files of greater taper resulted in a slight but detectable loss of apical root wall width. The



Fig. 12.13 Revascularization case illustrating treatment delivered to a 9-year-old male patient with a diagnosis of pulpal necrosis secondary to trauma, with a class 3 fracture in tooth #8 and a class 2 fracture in tooth #9 (**A**). The patient reported moderate to severe pain in both teeth. The teeth were isolated, accessed, and irrigated with 5% sodium hypochlorite, followed by placement of a mixture of ciprofloxacin, metronidazole, and minocycline for 55 days. Upon recall, the teeth were isolated, and the triple antibiotic paste was removed by irrigation. Bleeding was established in tooth #9 but not in tooth #8, where CollaCote was placed prior to mineral trioxide aggregate (MTA). The root canal systems were sealed with white MTA and a composite restoration (**B**). (Courtesy Dr. Alan Law.)

results indicated that revascularization treatment with either the TAP or $Ca(OH)_2$ medicament produced significantly greater increases in root length compared with either the MTA or NSRCT control groups. Treatment with the TAP produced significantly greater increases in root wall thickness compared with the MTA and NSRCT control groups. Treatment with $Ca(OH)_2$ resulted in significantly greater change in root wall thickness compared with the NSRCT group, but no differences were observed between these medicaments and the MTA apexification group. Finally, the TAP produced significantly greater differences in root wall thickness compared with either the $Ca(OH)_2$ or formocresol groups. In general, the formocresol group showed the smallest improvement in root length and wall thickness. Secondary analyses indicated that a 12- to 18-month recall is probably the minimal time to judge radiographic evidence of root development, although later time points (36 months) often demonstrate continued root

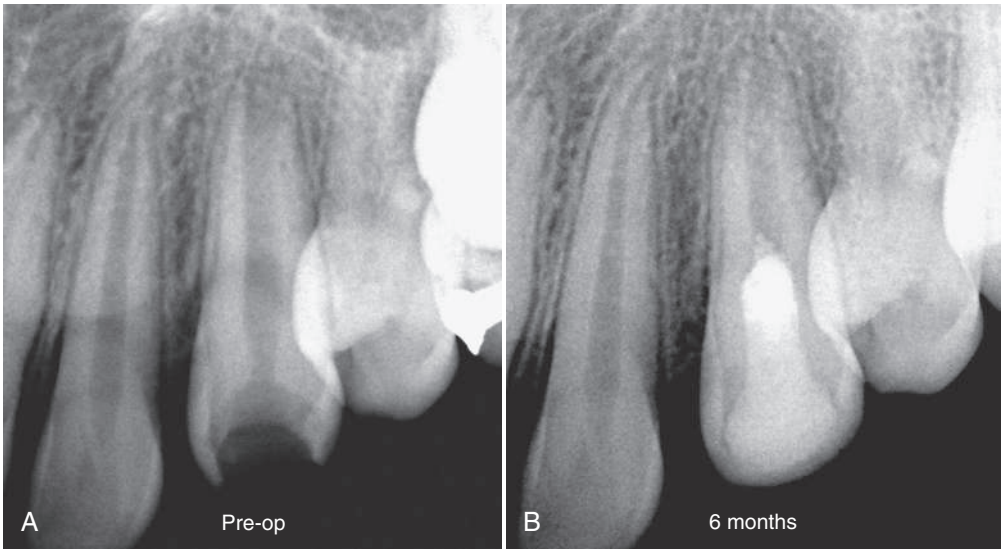


Fig. 12.14 Regenerative endodontic case illustrating treatment delivered to a 13-year-old female patient with a diagnosis of pulpal necrosis secondary to caries, with an unspecified prior history of trauma (**A**). The tooth was isolated, accessed, and irrigated with 5% sodium hypochlorite, followed by placement of a mixture of ciprofloxacin, metronidazole, and minocycline for 21 days. Upon recall, the tooth was isolated, and the triple antibiotic paste was removed by irrigation. Bleeding was established in both teeth, and the root canal system was sealed with white mineral trioxide aggregate and a composite restoration (**B**). (Courtesy Dr. Alan Law.)

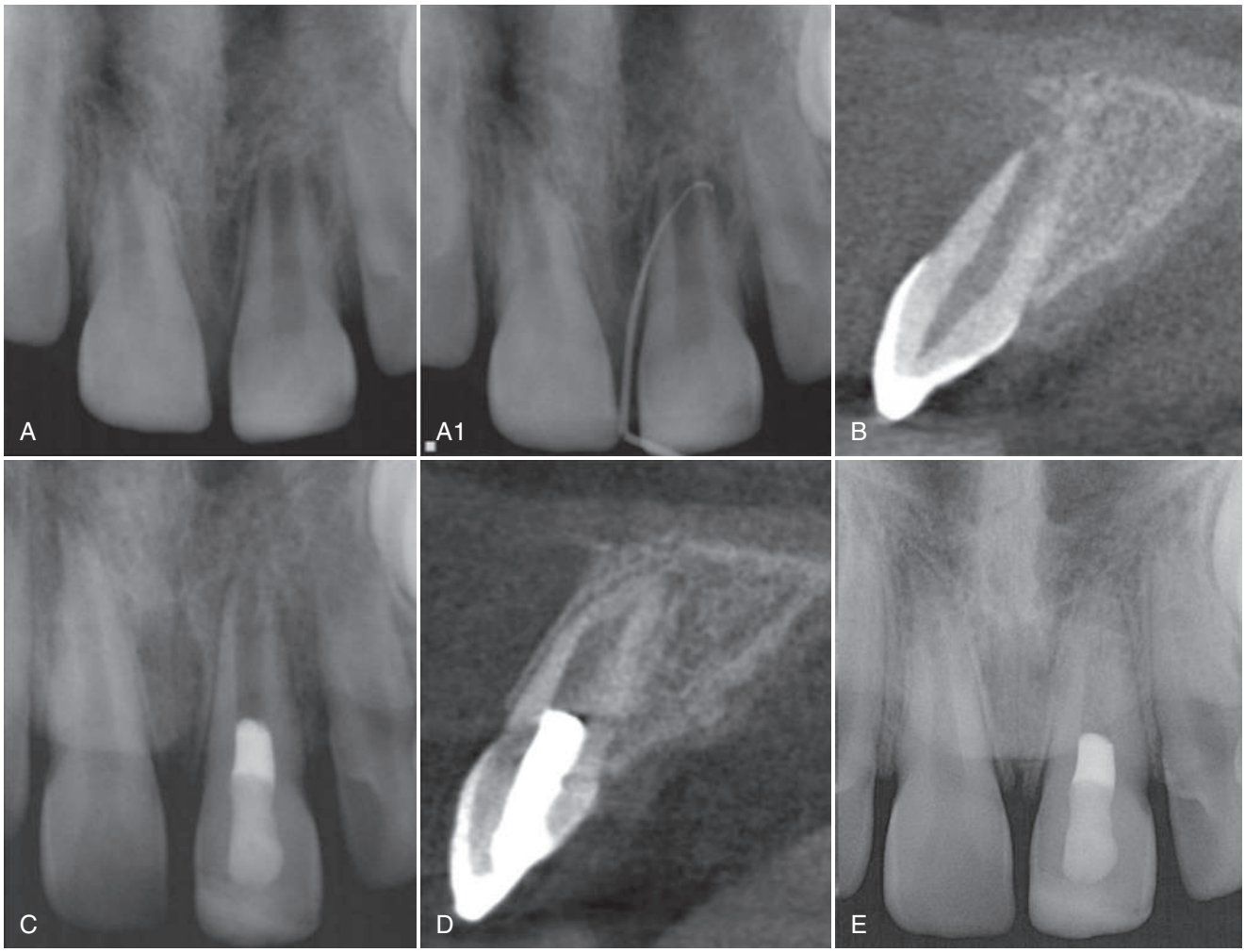


Fig. 12.15 Regenerative endodontic case illustrating treatment delivered to a 9-year-old male patient with a diagnosis of pulpal necrosis secondary to trauma and periradicular chronic apical abscess on tooth #9 seen on preoperative periapical radiograph (**A**) with sinus track tracing with a gutta-percha cone **(A1),** and with CBCT (**B**). The patient was asymptomatic and has draining facial sinus tract. The tooth was treated with a regenerative procedure using double antibiotic paste (DAP), a mixture of ciprofloxacin and metronidazole for 1 month. At the second appointment, the sinus tract had resolved. Then, the tooth was isolated, and the double antibiotic paste was removed by irrigation with 20 mL 17% EDTA. Intracanal bleeding was evoked from the apical tissues using a precurved #25 hand-file extending approximately 2 mm beyond the root apex. A Collaplug barrier was placed at the mid-root level, then covered with 3-mm white MTA. Fuji II LC glass ionomer was used as a coronal seal of the MTA. The access was then restored with a composite resin and polished (**C**). At the 1-year follow-up, the patient was asymptomatic, responsive to electrical pulp test (EPT), periodontal probings were no greater than 3 mm, and the tooth exhibited grade I mobility without coronal discoloration. Importantly there was appreciable root maturation in addition to resolution of apical radiolucency seen on CBCT (**D**) and periapical radiograph (**E**). (Courtesy of Dr. Obadah Austah, University of Texas Health Science Center at San Antonio.)

development. Jeeruphan et al. (2012) used a similar method to measure changes in root length and width following apexification versus revascularization.[152] They reported that teeth treated with revascularization showed a significantly greater percentage increase in root length (14.9%) compared with teeth treated by either MTA apexification (6.1%) or calcium hydroxide apexification. They also reported that the revascularization protocol produced significantly greater percentage increases in root width (28.2%) compared with teeth treated by either MTA apexification (0.0%) or calcium hydroxide apexification. In addition to measuring changes in root dimensions, the authors reported a survival (defined as retention of the tooth in the arch at the time of the postoperative recall) rate of 100% at an average of 14 months after revascularization treatment. This compared favorably with survival rates of 95% for MTA apexification and 77% for calcium hydroxide apexification. Although prospective randomized clinical trials with standardized radiographic assessment are clearly required, the results of this retrospective study are consistent with a robust outcome of revascularization procedures, particularly when medicated with either a TAP or a $Ca(OH)_2$ medicament.

EXAMPLE OF A REVASCULARIZATION PROTOCOL

Based on current research, several considerations can be reviewed when considering a protocol for revascularization treatment. The first issue is case selection; the best available evidence indicates that this treatment should be considered for the incompletely developed permanent tooth that has an open apex and is negative to pulpal responsiveness testing. Although the ultimate goal of this approach may

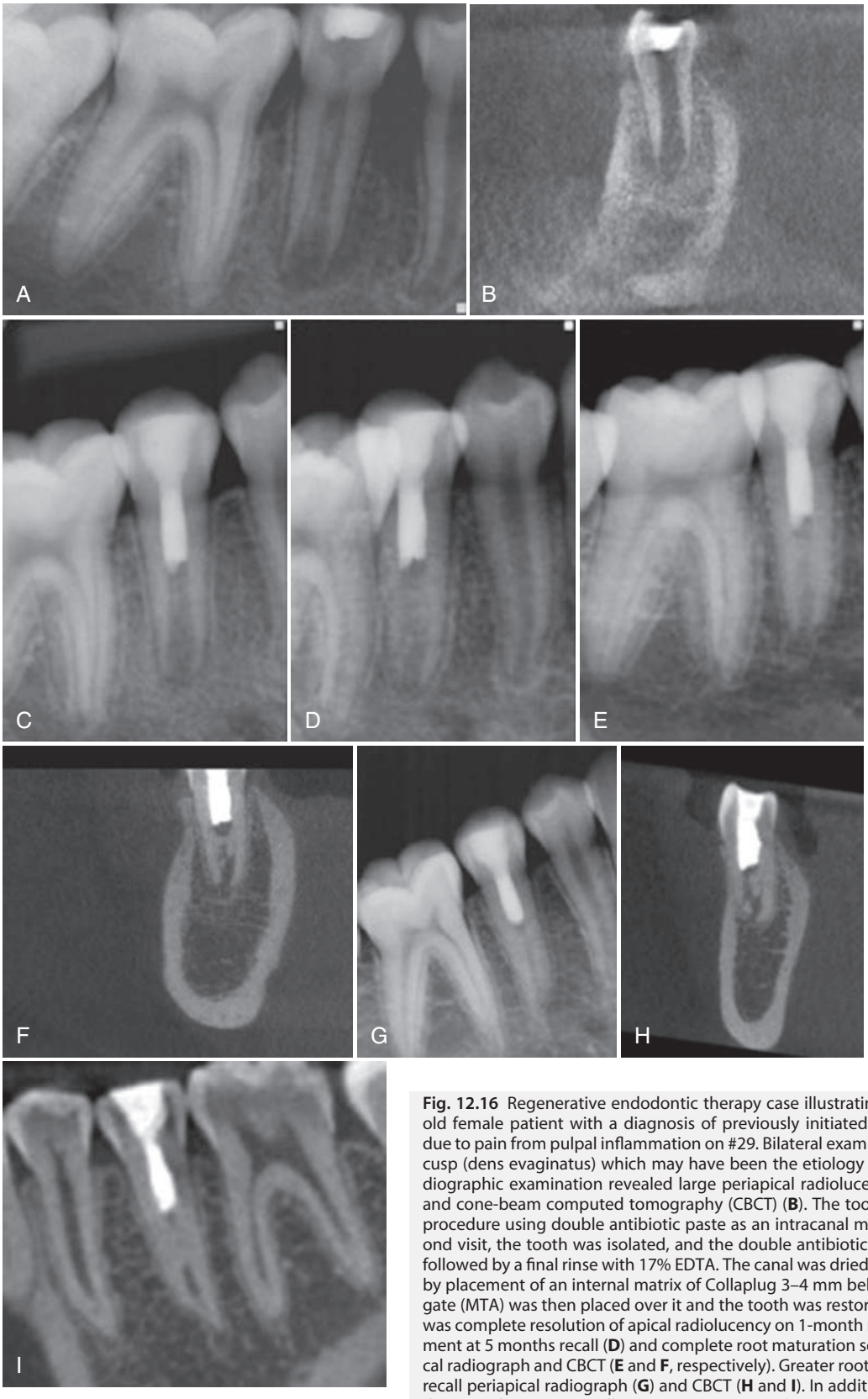


Fig. 12.16 Regenerative endodontic therapy case illustrating treatment delivered to a 9-year-old female patient with a diagnosis of previously initiated endodontic therapy (pulpectomy) due to pain from pulpal inflammation on #29. Bilateral examination revealed presence of a talon cusp (dens evaginatus) which may have been the etiology of pulpal inflammation on #29. Radiographic examination revealed large periapical radiolucency on a periapical radiograph (**A**) and cone-beam computed tomography (CBCT) (**B**). The tooth was treated with a regenerative procedure using double antibiotic paste as an intracanal medicament for 48 days. On the second visit, the tooth was isolated, and the double antibiotic paste removed by saline irrigation followed by a final rinse with 17% EDTA. The canal was dried and bleeding was evoked followed by placement of an internal matrix of Collaplug 3–4 mm below the CEJ. Mineral trioxide aggregate (MTA) was then placed over it and the tooth was restored with Fuji II LC and Build It. There was complete resolution of apical radiolucency on 1-month recall (**C**), appreciable root development at 5 months recall (**D**) and complete root maturation seen at 1-year recall on both periapical radiograph and CBCT (**E** and **F**, respectively). Greater root maturation is observed at 2.5 years recall periapical radiograph (**G**) and CBCT (**H** and **I**). In addition, the tooth responded to EPT on the 1- and 2.5-year recall visits. (Courtesy Dr. Nikita Ruparel, University of Texas Health Science Center at San Antonio.)

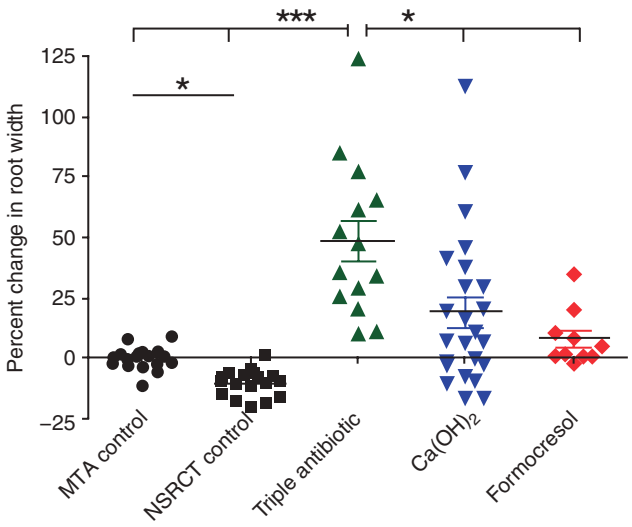


Fig. 12.17 Retrospective analysis of the percentage of change in dentinal wall thickness from preoperative image to postoperative image, measured at the apical third of the root (position of apical third defined in the preoperative image) in 40 control patients and in 48 patients following a revascularization procedure. *, $P < .05$ for each $P < .05$; ***, $P < .001$ versus mineral trioxide aggregate (MTA) apexification control group and NSRCT control group. $P < .05$ versus NSRCT control group only. $P < .05$ versus calcium hydroxide ($Ca[OH]_2$) and formocresol groups. $P < .05$ versus NSRCT control group only. (From Bose R, Nummikoski P, Hargreaves K: A retrospective evaluation of radiographic outcomes in immature teeth with necrotic root canal systems treated with regenerative endodontic procedures, *J Endod* 35:1343–1349, 2009.)

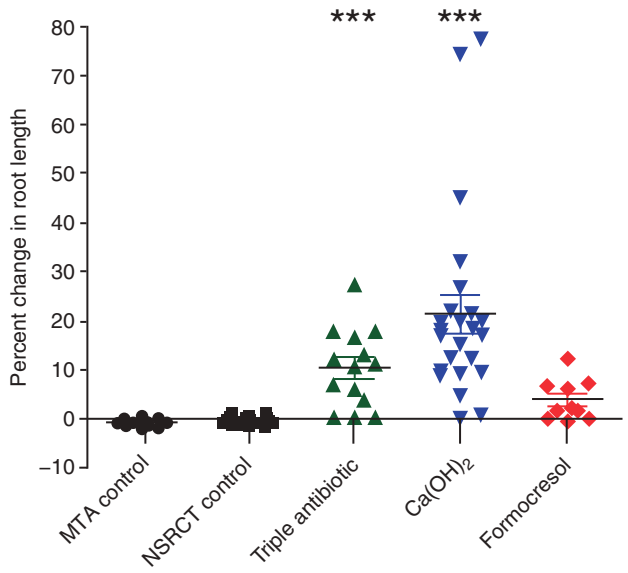


Fig. 12.18 Retrospective analysis of the percentage of change in root length from preoperative image to postoperative image, measured from the cementoenamel junction (CEJ) to the root apex in 40 control patients and in 48 patients following a revascularization procedure. ***$P < .001$ versus mineral trioxide aggregate (MTA) apexification control group ($n = 20$) and NSRCT control group ($n = 20$). $P < .05$ versus MTA control group only. Median values for each group are depicted by horizontal line, and individual cases are indicated by the corresponding symbol. (From Bose R, Nummikoski P, Hargreaves K: A retrospective evaluation of radiographic outcomes in immature teeth with necrotic root canal systems treated with regenerative endodontic procedures, *J Endod* 35:1343–1349, 2009.)

include a tissue engineering–based method of pulpal regeneration in the fully developed permanent tooth, it should be recognized that current revascularization protocols have not been developed or evaluated for these more challenging cases. Informed consent should include the number of appointments (at least two), the possible adverse effects (primarily potential staining of the crown), the potential lack of response to treatment and alternative treatments, and possible posttreatment symptoms. Because the canal space will not be accessible following revascularization, teeth requiring retention in the canal space for the restoration are not good candidates for REPs. One reason for clinical staining of the crown and any root structure above the gingival margin appears to be due to the presence of minocycline.[169,235,241] This can be minimized by using a delivery system that restricts the drug below the cementoenamel junction (CEJ)[241] or the use of medication that does not contain minocycline such as double antibiotic mixture (ciprofloxacin and metronidazole) or calcium hydroxide. The use of MTA (ProRoot, Dentsply Tulsa Dental, Tulsa, OK), in both gray- and white-form variants,[33,43,48] may also cause tooth discoloration, which can also be reduced but not completely eliminated by a walking bleach technique.[43] Importantly, newer tricalcium silicate materials such as Biodentine and Endosequence BC Root Repair Material-Putty (ES) have substantially less discoloration potential, while promoting greater proliferation of SCAPs and differentiation into odontoblastic-like cells than MTA in an in vitro regenerative endodontics model.[200] This is particularly important because these nonstaining materials can be placed above the CEJ, favoring the formation of mineralized tissues in the cervical region of teeth, where ferrule and resistance to fracture are important factors. Importantly, because MTA promoted greater osteoblastic potential than the other tested biomaterials, it is recognized as the best-suited material to be used in MTA plugs and endodontic microsurgeries when bone formation and repair is the desired outcome.[200] Alternative treatments that should be discussed with the patient and guardian would include MTA apexification, no treatment, or extraction.

At the first appointment (Fig. 12.19, *A–E*), the treatment alternatives, risks, and potential benefits should be described to the patient and guardian after collecting clinical information and establishing pulpal and apical diagnoses. Following informed consent, the tooth is anesthetized, isolated, and accessed. Minimal instrumentation should be accomplished perhaps with the use of Hedstrom files circumferentially "brushing" the canal walls without major dentin removal. In addition, an appropriately sized file should be placed in the root canal system and radiographed to determine working length is important. Next, the root canal system should be copiously and slowly irrigated. Because of its proven efficacy as a canal disinfectant and a tissue dissolution agent, sodium hypochlorite (NaOCl) has been the irrigant of choice in most the revascularization case reports.[77] However, studies have demonstrated that treatment of dentin with NaOCl results in an indirect cytotoxic effect on stem cells.[194,300] Importantly, minimal deleterious effect was seen with the use of 1.5% NaOCl followed by 17% EDTA. Thus lower concentration (1.5%) of NaOCl followed by EDTA should be considered as the standard irrigation for REPs.[194] Moreover, the use of chlorhexidine should be limited or avoided because it does not have tissue

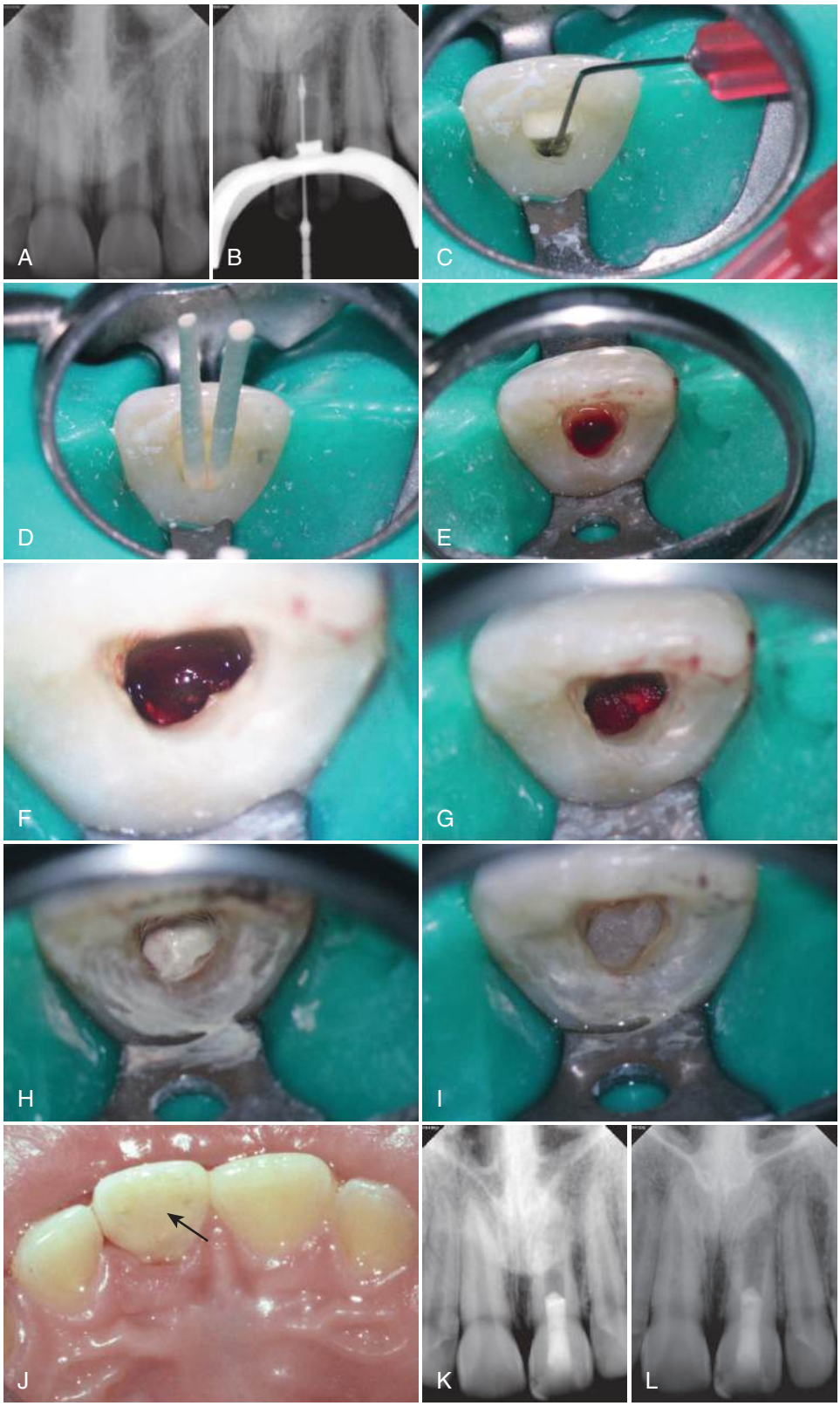


Fig. 12.19 Example of a revascularization case. Treatment was delivered to a 12-year-old boy who suffered trauma on tooth#9 2 years prior to the appointment. First appointment: (**A**), Clinical examination revealed a pain upon percussion and palpation. A working length file was placed into the root canal system (**B**). The tooth was slowly irrigated with 20 mL of 1.5% sodium hypochlorite (NaOCl) followed by 20 mL of saline using a Max-I-Probe needle inserted to the apical third (**C**). The canals were dried and medicated with calcium hydroxide paste (Ultracal) (**D**). The patient returned asymptomatic 1 month later. The tooth was isolated, accessed, and the medicament removed by slow irrigation with 1.5% NaOCl followed by 17% EDTA. After the canal was adequately dried, intracanal bleeding was achieved by lacerating the apical tissues (**E**). Blood clotting was observed after approximately 10 minutes **(F).** CollaPlug was placed below the CEJ (**G**) to serve as a matrix to position the white mineral trioxide aggregate coronal to the blood clot (**H**). The tooth was then sealed with a layer of Fuji IX (**I**), etched and restored with a composite (**J**). A final radiograph was taken (**K**). The tooth responded to electrical pulp tested at the 1-year recall, and demonstrated apical closure and moderate dentinal wall thickening (**L**). (Courtesy Dr. Anibal Diogenes, University of Texas Health Science Center at San Antonio.)

dissolution capability and has also been shown to be cytotoxic to stem cells. However, if chlorhexidine is used, its cytotoxicity can be largely avoided by adding an inactivation step with a solution of L-alpha-lecithin followed by 17% EDTA.[310] Because canal disinfection relies considerably on chemical irrigants, it is important to place the needle into the apical third and irrigate using needles with a closed end and side-port vents (e.g., Max-I-Probe needles), together with a slow rate of infusion, to help to reduce any irrigants passing through the open apex. There is evidence that negative pressure can be beneficial and it should be considered in these procedures.[70] In addition, ultrasonic activation of solutions has been shown to promote greater tissue dissolution in addition to accelerating the release of growth factors from the dentinal walls.[312] Thus ultrasonic activation should be considered, particularly in the last visit for both sodium hypochlorite and the final EDTA irrigation.

The root canal system is then dried with sterile paper points, and the antimicrobial medicament is delivered into the root canal space. The best available evidence would support the use of either an antibiotic mixture (TAP or double antibiotic paste [DAP]) at the concentration of 1 to 10 mg/mL or $Ca(OH)_2$. TAP and $Ca(OH)_2$ have been shown to be effective (see Figs. 12.13 and 12.14). The TAP has the advantage of being a very effective antibiotic combination against endodontic microorganisms; its efficacy is supported by its use in many published cases.[77,173] However, this combination is not approved by the FDA and carries a potential for minocycline staining of the crown. Thus, when performing REPs on teeth in esthetic areas, practitioners should consider the use of calcium hydroxide or eliminating minocycline from the antibiotic paste, sealing the coronal dentin with a dentin-bonding agent or composite. After antimicrobial medicament is placed, the tooth is then sealed with a sterile sponge and a temporary filling (e.g., Cavit), and the patient is discharged for 3 to 4 weeks.

At the second appointment (see Fig. 12.19, *F–M*), the patient is evaluated for resolution of any signs or symptoms of an acute infection (e.g., swelling, sinus tract pain) that may have been present at the first appointment. The antimicrobial treatment is repeated if resolution has not occurred. In most reported cases, the acute signs and symptoms resolved after treatment with the intracanal medicament.[77] Because revascularization-induced bleeding will be evoked at this appointment, the tooth should not be anesthetized with a local anesthetic containing a vasoconstrictor. Instead, 3% mepivacaine can be used, which will facilitate the ability to trigger bleeding into the root canal system. Following isolation and reestablishment of coronal access, the tooth should be copiously and slowly irrigated, possibly together with gentle agitation with a small hand file to remove the antimicrobial medicament, followed by ultrasonic activation of the irrigant with the tip 2 to 3 mm away from the apical foramen. When choosing an irrigant at the second appointment, it is worth considering irrigants such as NaOCl and chlorhexidine may be cytotoxic to stem cells either directly[89] or indirectly after dentin has been exposed to the irrigants.[194,300,310] It has been demonstrated that exposure of dentin to 5% to 6% sodium hypochlorite leads to decreased stem cell survival and odontoblastic differentiation. This indirect effect is likely related to various deleterious effects of sodium hypochlorite on the dentin matrix leading to reduced cell attachment[242] and a decrease in dentin matrix-derived growth factors such as TGF-β1.[270,332] Thus it is prudent to continue to use sodium hypochlorite at the concentration of 1.5% on the second visit. Lastly, irrigation with 17% EDTA may be advantageous because it has been shown to release growth factors from dentin and promote stem cell survival and differentiation.[312,332] In addition, Galler et al. (2011), using dentin cylinders transplanted into immunocompromised mice, showed that DPSCs seeded onto EDTA-conditioned dentin differentiated into odontoblast-like cells and had cellular processes extending into the dentin.[106] On the other hand, stem cells differentiated into osteoclasts/odontoclasts, causing resorption on the dentinal walls if dentin was previously treated with 5% sodium hypochlorite. Thus the use of 17% EDTA as the last irrigant promotes attachment, proliferation, and odontoblastic differentiation of stem cells that could further increase with the use of passive ultrasonic irrigation.[312]

After drying the canal system with sterile paper points, a file is placed a few millimeters beyond the apical foramen, and the apical tissue is lacerated with bleeding preferably coronal to the CEJ. A small piece of Colla-Plug (Zimmer Dental, Carlsbad, CA) may be placed over the clot to serve as an internal matrix and prevent the restorative materials from being displaced into the canal space upon placement. Approximately 2 to 3 mm of a bioactive hydraulic cement such as Biodentine, ES, or MTA should be placed as coronal as possible while leaving enough chamber space for a glass ionomer and composite restoration. Although MTA has been used in many of the case reports and may have many desirable properties,[230,231,294,298] it is associated with severe discoloration, making it unsuitable to be used in these procedures.[160,172,174,180] Alternatively, Biodentine or ES putty could be used because there have been encouraging studies demonstrating that they induce stem cell proliferation and odontoblastic differentiation.[178,192,200,328] Indeed, Biodentine[11,282,293] and ES putty[45] have been successfully used in REPs and appear better suited materials due to the reduced risk of discoloration.

In addition to follow-up visits in the first several months after REPs to assess for signs of healing, a 12- to 18-month recall should be considered as a reasonable time point to radiographically assess apical healing and root development.[39]

The following box summarizes the steps in a REP. Importantly, these considerations are based on the best available evidence and are likely to change as the field progresses (Box 12.1).

CLINICAL MEASURES OF TREATMENT OUTCOMES

The goal of conventional endodontic therapy is to maintain or restore the health of periradicular tissues by preventing or healing apical periodontitis. The goals of regenerative procedures extend beyond the goals of conventional endodontic therapy and include continued root development and reestablishment of pulpal vitality while aiming to achieve the functional recapitulation of a healthy pulp-dentin complex. As mentioned earlier, contemporary REPs are being modified and evolving towards accomplishing

Box 12.1 Treatment Procedures for Regenerative Endodontics

First Treatment Visit for Regenerative Endodontics:

1. Informed consent, including explanation of risks and alternative treatments or no treatment.
2. After ascertaining adequate local anesthesia rubber dam isolation is obtained.
3. The root canal systems are accessed and working length is determined (radiograph of a file loosely positioned at 1 mm from root end).
4. The root canal systems are slowly irrigated first with 1.5% NaOCl (20 mL/canal, 5 min) and then irrigated with saline (20 mL/canal, 5 min), with irrigating needle positioned approximately 1 mm from root end.
5. Canals are dried with paper points.
6. Calcium hydroxide or an Antibiotic Paste or solution (combined total of 1–10 mg/mL) is delivered to canal system
7. Access is temporarily restored

Final (Second) Treatment Visit for Regenerative Endodontics

(Typically 2–4 Weeks After the First Visit):

1. A clinical exam is first performed to ensure that that there is no moderate to severe sensitivity to palpation and percussion. If such sensitivity is observed, or a sinus tract or swelling is noted, then the treatment provided at the first visit is repeated.
2. After ascertaining adequate local anesthesia with 3% mepivacaine (no epinephrine) rubber dam isolation is obtained.
3. The root canal systems are accessed; the intracanal medicament is removed by irrigating with 17% ethylenediaminetetraacetic acid (EDTA) (30 mL/canal, 5 min) and then a final flush with saline (5 mL/canal, 1 min).
4. The canals are dried with paper points.
5. Bleeding is induced by rotating a precurved K-file size #25 at 2 mm past the apical foramen with the goal of having the whole canal filled with blood at least to the level of the cementoenamel junction, but preferably coronal to it.
6. Once a blood clot is formed, a premeasured piece of Collaplug (Zimmer Dental Inc., Warsaw, IN) is carefully placed on top of the blood clot, preferably above the cementoenamel junction, to serve as an internal matrix for the placement of approximately 2 mm of a tricalcium silicate biomaterial such as Biodentine (Septodont) or Endosequence BC Root Repair Material-Putty (ES) (Brasseler).
7. A (3 mm) layer of glass ionomer layer (e.g., Fuji IX, GC America, Alsip, IL; or other) is flowed gently over the bioactive coronal barrier and light cured for 40 s.
8. A bonded reinforced composite resin restoration is placed over the glass ionomer.
9. The case needs to be followed-up at 3 months, 6 months, and yearly after that for a total of 4 years.

these additional objectives that are not achievable with conventional nonsurgical root canal therapy. Accordingly, the definition of success for REPs includes multilevel goals that are not mutually exclusive. This multilevel approach should be focused on three levels of outcomes: patient-centered, clinician-centered, and scientist-centered outcomes (Fig. 12.20).

Patient-Centered Outcomes

It is well established that the primary goal of endodontics is the prevention and healing of pulpal and periradicular inflammation (i.e., resolution of signs and symptoms of disease). Unfortunately, there has been too much emphasis on radiographic assessment while often ignoring the benefit and improvement in the quality of life given to the patient. Clinical studies focusing on patient-centered outcomes in endodontics report that the greatest benefit of endodontic therapy was alleviation of pain and psychological discomfort[202] and improvement in quality of life.[129] Unfortunately, a tooth could be considered a failed treatment if any radiographic finding is detected by a clinician despite the benefit that the treatment has brought to the patient. Not surprising, the definition of clinical success has been a controversial topic in endodontics throughout the history of the specialty, but it has been largely based on lack of symptoms and a strict radiographic criteria,[34,35,102,281] or even dependent on the absence of cultivable microorganisms.[47] This radiographic criteria of success have been present in endodontics for decades and exemplify the disconnect between the expectations of patients, clinicians, and scientists. This disconnect is not unique to endodontics but found in all fields of dentistry and medicine. Patients are the primary stakeholders of their own well-being status, providing feedback on treatment effectiveness, and thus advancing the development of therapies that promote the overall population health. For patients undergoing regenerative procedures, their primary expectation is that teeth are disease free (i.e., no symptoms) and functional (i.e., survival) and have acceptable esthetics.[78]

A recent metanalysis of published REPs and apexification procedures revealed that the success defined as healing of apical periodontitis was found in approximately 91% and 95% for REPs and apexification procedures without statistically significant difference between the two treatment modalities.[297] Likewise, there was no difference in survival rate between REPs and apexification procedures, 98% and 97%, respectively. As pointed out in this metanalysis, most included studies were case series and retrospective studies. Additional randomized clinical trials with large sample sizes and adequate follow-up (>2 years) and recall rates are required to improve the quality of the pooled data. It is important to highlight that this lack of higher-quality studies is not restricted to REPs, but it is also true for apexification procedures. Therefore so far there have not been studies that demonstrated a meaningful difference between REPs and apexification in healing of apical periodontitis and survival rates, with the latter impacted by the often-found short follow-up periods in the current published studies. Collectively, currently published studies support that both procedures demonstrate equally acceptable clinical outcomes for healing and survival addressing two of the three patient-centered outcomes.

Esthetics is considered a patient outcome because it is a personal perception. Tooth discoloration has been identified as an adverse event associated with many published REPs.[160,218,235] Tooth discoloration has two primary sources in these procedures: intracanal medicament and coronal barrier material. As discussed previously in this chapter, the undesirable staining that has marked the initial generations of these procedures can be greatly minimized, or completely eliminated, with the use of medicaments that do not contain minocycline and alternative tricalcium silicate biomaterial with lesser staining risk than MTA, such as

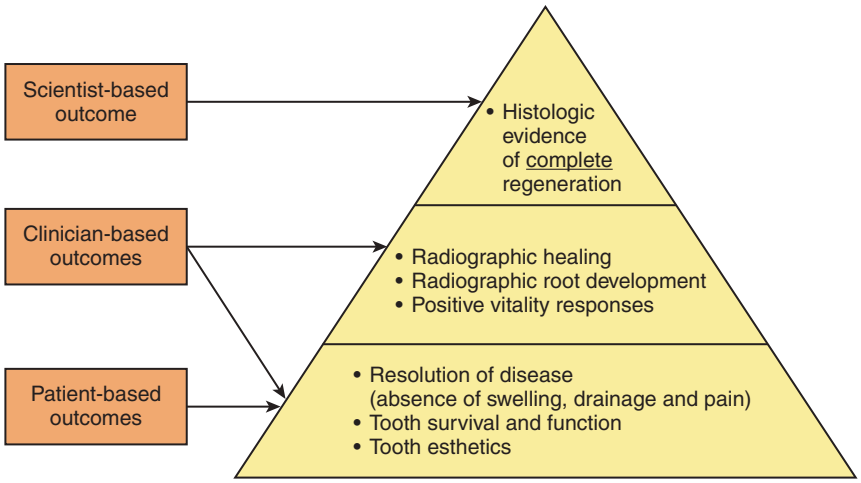


Fig. 12.20 Trilevel outcome assessment pyramid. Regenerative endodontic procedures must be assessed and evaluated in a systematic manner, acknowledging that there are different levels of possible outcomes that carry different meanings and value for patients, clinicians, and scientists. The patient-centered outcomes represent the base of this pyramid, symbolizing the fundamental importance of achieving healing, reestablishment of function, and patient satisfaction and overall well-being. (Modified from Diogenes A, Ruparel NB, Shiloah Y, Hargreaves KM: Regenerative endodontics: a way forward, *JADA* 147(5):372–380, 2016.)

Biodentine or ES, in esthetic zones.[49,167] In addition, the use of a dentin-bonding agent to seal the dentinal tubules in the pulp chamber is beneficial also to minimizing any possible discoloration from the coronal plug material.[6,169] Therefore a clinician's choice of medicaments and materials has the potential to greatly impact this important patient-centered outcome.

Clinician-Centered Outcomes

Because the well-being of our patient is the paramount goal of every clinician, we must first ascertain that the patient-centered outcomes (i.e., no symptoms, functional survival, and acceptable esthetics). In addition, clinicians use objective and subjective clinical testing to assess radiographic evidence of healing and continued root development and responses to sensibility tests (i.e., vitality testing).

Continued root development is another desirable outcome of REPs. This is normally evaluated by the clinician upon radiographic evaluation. A total of four possible radiographic outcomes regarding root development can be found (Fig. 12.21). These outcomes range from: type 4—no change; type 3—closing of apical opening without change in elongation/thickening; type 2—root elongation/thickening without closure of the apical opening; and type 1—elongation/thickening and closure of the apex (most desirable outcome). Many published case reports and case series provided initial subjective evidence that REPs promoted radiographic healing and continued root development, until a retrospective study used a quantitative methodology to assess root development, comparing to apexification procedures. In this initial study, REPs were found to promote significantly greater root elongation and thickening compared with no changes seen with apexification procedures.[39] The methodology used in this study was adopted by other studies finding similar results.[13,153,159,211] Furthermore, this methodology was later revised and modified to assess changes in radiographic root area.[97] Despite being designed to digitally align radiographs taken in a nonstandardized manner to allow for quantitative measurements, there are instances in which the discrepancy in angulation and acquired image dimension ratios exceed the capacity of the software to

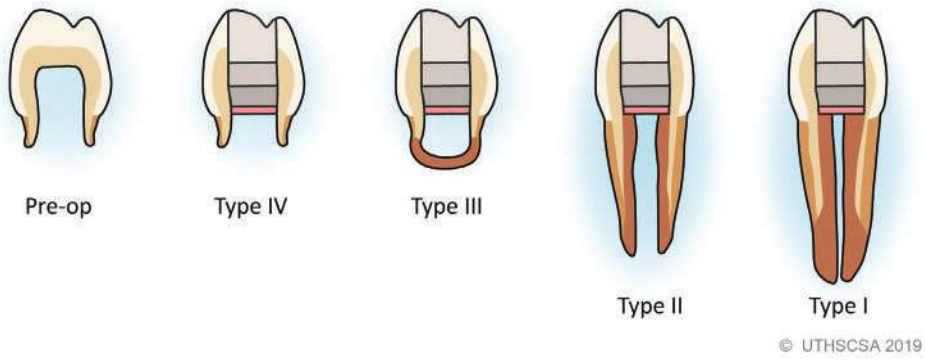


Fig. 12.21 Schematic illustrating the different possible outcomes in root development following regenerative endodontic procedures. Type I shows root development and narrowing of the apical foramen. Type II shows root development but failure of apex closure. Type III shows no root development but closure of the apical foramen. Type IV shows no developments and failure to apex closure.

perform adequate correction.[159] As a viable alternative, volumetric analysis of acquired cone-beam computed tomography (CBCT) images has been proposed[91] as a method to three-dimensionally quantify changes in root development. This methodology has been used in two studies, both demonstrating that REPs promoted an increase in total root volume.[24,198] A recent randomized prospective study used acquired CBCT volumes to quantify and characterize the type of development detected in cases treated with REPs ($n = 69$) and apexification procedures ($n = 34$). In this study, REPs promoted the average root growth of greater than 0.24 mm in approximately 82.6% of the teeth, whereas teeth treated with apexification had no changes in root development. Interestingly, teeth with dental anomaly (i.e., dens invaginatus) as etiology of disease showed more desirable outcome patterns (type 1 and type 2; see Fig. 12.21), whereas trauma was associated with less development and greater incidence of the less desirable type 4 and type 3 outcome patterns (see Fig. 12.21). These data agree with previous case series and retrospective studies that included trauma as the main etiology for less desirable outcomes[13,257] compared with other studies that included less trauma cases.[153] Therefore these studies suggest that trauma negatively affects root development following REPs. Even though more studies are required to establish the correlation of trauma with less desirable outcomes in REPs, these observations agree with different lines of evidence demonstrating that development can be halted if trauma disrupts the interaction between the apical papilla and the HERS.

Vitality responses have been reported in approximately 50% of the published cases.[77] Responses to electric pulp tester (EPT) are more commonly reported than cold responses. These responses to vitality testing (with either cold or EPT), as well as the lack of signs and symptoms of pathosis, suggest the presence of functioning tissue in the canal space. Vitality responses, in addition to continued root development, are a very desirable outcome because "normal" nociception is protective and suggests immune competence of the vascularized tissue. However, the lack of response to sensitivity tests in absence of signs and symptoms of disease should not be interpreted as failure, because there have been several reports of cases demonstrating healing of apical periodontitis and appreciable root development in the absence of a positive response to vitality testing. Histologic examination of teeth treated with REPs has revealed the presence of nerve fibers in a well-vascularized loose connective tissue.[179,199] A recent histologic study of a tooth treated successfully with REPs revealed that this innervation includes nociceptors, thus lending evidence that functional free nerve endings are regenerated following these procedures.[24] These free nerve endings are attracted to the "regenerated" pulp space by a process called axonal targeting that is likely mediated by cellular and noncellular processes.[25,72,88]

The significance of reinnervation is the newly formed tissues within the canal space likely transcend the expected physiologic responses to noxious stimuli and infections. It is known that several species of amphibians including salamanders have the capacity of full limb regeneration. Interesting, this amazing regenerative capacity, which has been studied by researchers for decades, is completely lost if the limb is denervated.[92] This concurrent loss of innervation and regenerative potential suggests that peripheral nerves have a pivotal role in regeneration. Further research is required to establish regenerative role in the pulp-dentin complex. Nonetheless, the appropriate innervation of the newly formed pulplike tissue must represent a major goal of these procedures. Lastly, the presence of reinnervation may not always be detected with sensitivity tests that are prone to false-positive and false-negative responses[182] in addition to variations in the thickness and apical extent of coronal barrier and restorations.

Scientist-Based Outcomes: Regeneration or Repair?

Scientific outcomes that are not directly related to the well-being of the patient or the clinical success according to clinician-based outcome measures can be of high importance. These outcomes require scientific methodology to be evaluated and focus on important scientific questions, promoting further research and bring advancements to the field. A good example of a scientist-based outcome is the nature of the tissue formed, including the identification of cellular components, formed following REPs. Undoubtedly, there is considerable debate on whether REPs promote true regeneration or repair.[268] This has strong scientific merit, but it is directly related to the clinical success of a case. Histologic evaluation of teeth that were previously treated with REPs but later extracted due to recurring trauma and fractures suggests that the newly formed tissue does not resemble the lost dental pulp, and formed tissues include ectopic bone, cementum, and periodontal ligament, in addition to commonly found components of the dental pulp complex such as fibroblasts, blood vessels, and nerves.[30,195,199,264]

A recent histologic study evaluated the scientist-based outcomes of two REP cases with different clinical outcomes.[24] One case was considered a clinical failure due to recurrent caries and resorption, despite the noticed continued root development resulting in extraction and histologic examination, which revealed the presence of inflammatory infiltrate absence of odontoblast-like cells and the predominant presence of cementum and bone as newly mineralized tissues. The other case was considered a clinical success for 6 years and was extracted only for orthodontic reasons. In addition to the typical ectopic mineralized tissues, odontoblast-like cells were found and newly formed tubular dentin (Fig. 12.22).[24] In addition, this study characterized the innervation found in the tissue as containing peptidergic nociceptors (i.e., expressing calcitonin gene related peptide) (Fig. 12.23). This is the first study that used laser confocal microscopy to identify cellular components and mineralized matrix in teeth treated with REPs.[24] The results of this study are surprising and suggest that, when sophisticated methods are used, further characterization can be achieved, revealing that at least a partial regeneration of components of the pulp-dentin complex is possible. It has been suggested that the current scientist-based status of REPs could be described as guided endodontic repair (GER) because the tissues are not fully regenerated but demonstrate the result of an intentional and directed repair that was not previously thought possible.

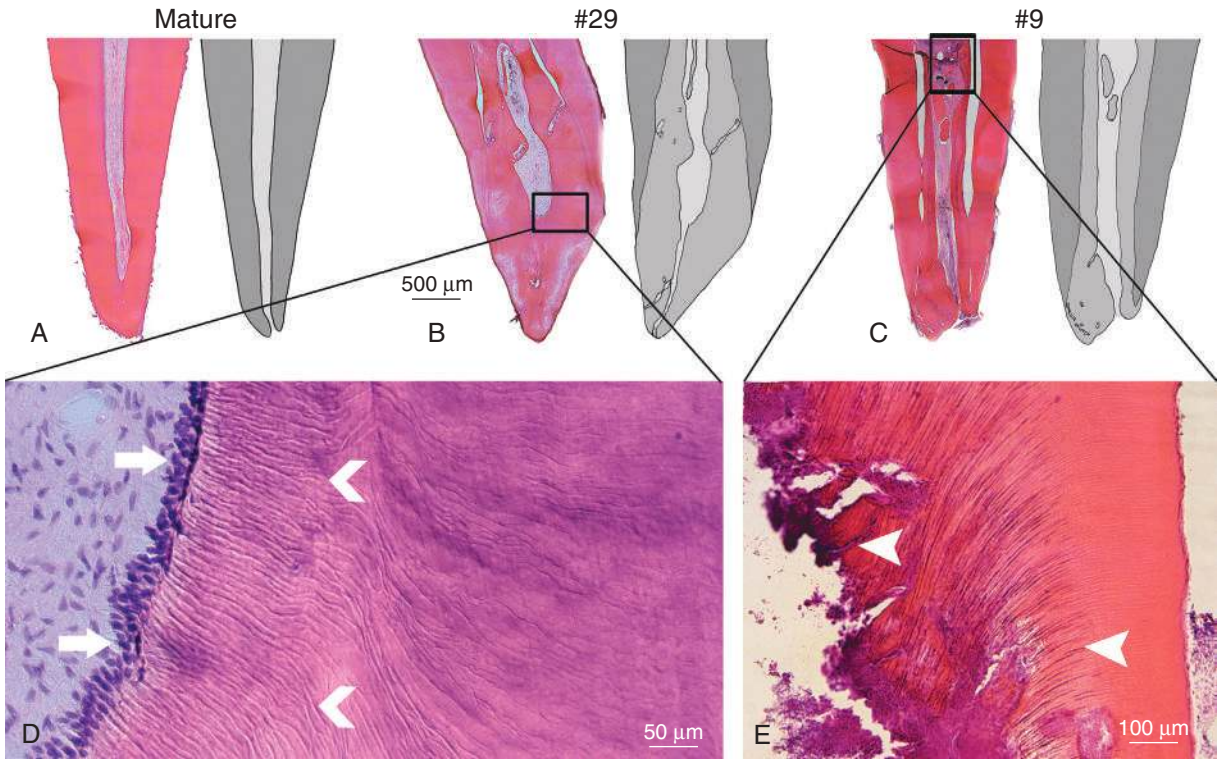


Fig. 12.22 (**A–C**) Representative hematoxylin-eosin staining of longitudinal sections of a healthy mature tooth #28, tooth #29, and tooth #9. (**D**) An enlarged section from **B** demonstrating mature, secretory odontoblasts lining newly formed tubular dentin *(arrows)*. (**E**) An enlarged section from **C** showing the presence of biofilm colonies attached and penetrated into dentinal tubules of tooth #9 *(arrows)*. (From Austah O, Joon R, Fath WM, et al: Comprehensive characterization of 2 immature teeth treated with regenerative endodontic procedures, *J Endod* 44(12):1802–1811, 2018.)

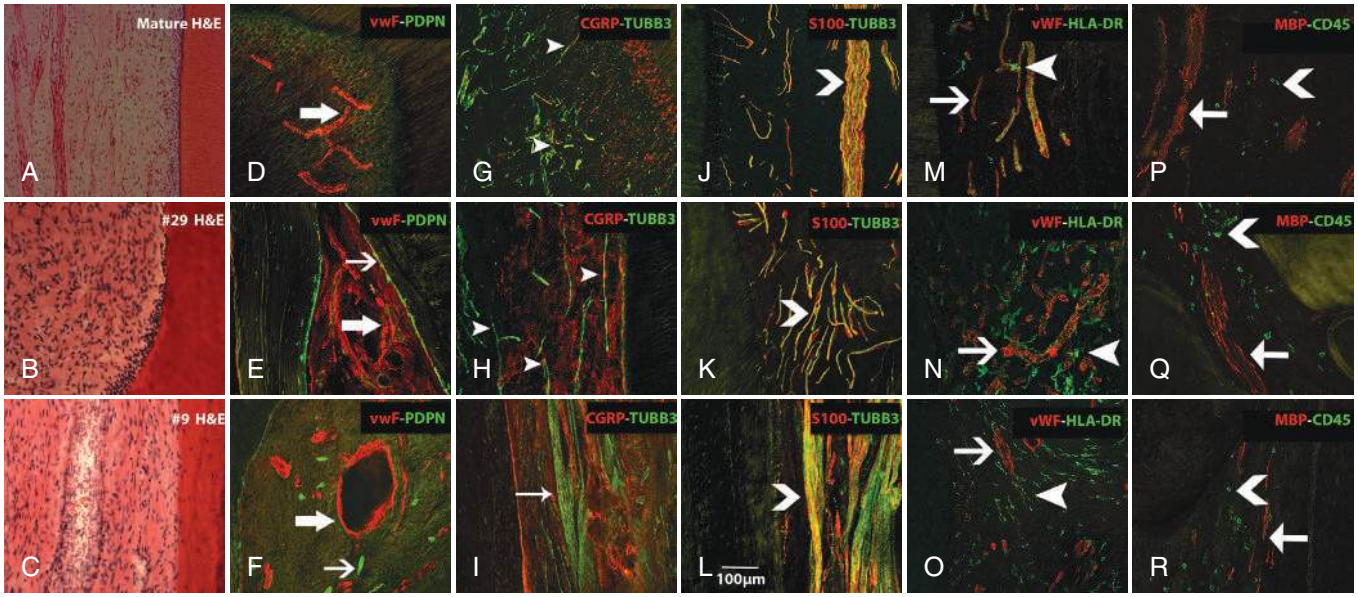


Fig. 12.23 (**A–C**) Representative hematoxylin-eosin staining of sections of a healthy mature tooth #28, tooth #29, and tooth #9. (**D–F**) Immunohistochemical staining showing von Willebrand factor and podoplanin (PDPN) immunoreactivity in a healthy mature tooth #28, tooth #29 and tooth #9. (**G–I**) Immunohistochemical staining showing colocalization of calcitonin gene–related peptide (CGRP) and class III beta-tubulin (TUBB3) in a healthy mature tooth #28 and tooth #29. (I) Tooth #9 shows TUBB3 immunoreactivity but no colocalization with calcitonin gene–related peptide. (**J–L**) Immunohistochemical staining demonstrating colocalization of Schwan cell marker (S100) with TUBB3 in a healthy mature tooth #28, tooth #29, and tooth #9. (**M–O**) Immunohistochemical staining showing close proximity of blood vessels stained with von Willebrand factor and a marker for antigen-presenting cells (human leukocyte antigen–antigen D related; HLA-DR)) in a healthy mature tooth #28, tooth #29, and tooth #9. (**P–R**) Immunohistochemical staining demonstrating close proximity of blood vessels stained with von Willebrand factor and a pan-leukocytic marker (CD45) in a healthy mature tooth, tooth #29, and tooth #9. *Arrows* indicate positive expression of the various proteins. (From Austah O, Joon R, Fath WM, et al: Comprehensive characterization of 2 immature teeth treated with regenerative endodontic procedures, *J Endod* 44(12):1802–1811, 2018.)

Summary

The field of regenerative endodontics is rapidly advancing. This progress is based on the principles of tissue engineering, namely the spatial delivery of appropriate cells, scaffolds, and growth factors. Similar to most rapidly developing fields, the preclinical area of research has outpaced translational clinical studies. Preclinical studies have identified several MSC sources capable of differentiating into odontoblast-like cells, as well as candidate scaffolds and growth factors capable of guiding this development. The initial preclinical animal studies indicate that using all three components of the tissue-engineering triad (stem cells, growth factors, and scaffolds) can result in complete regeneration of the pulp-dentin complex. There is a delicate balance between disinfection of the root canal system and the interplay of these three components that warrants more research. It is likely that this combined approach of in vitro and in vivo preclinical research will greatly advance our understanding of the conditions necessary to regenerate a functional pulp-dentin complex.

The translational nature of regenerative endodontic research is allowing for changes to take place in clinic practice in a relatively short time. This crosstalk between basic and clinical sciences is largely fueled by the realization that all three components of the tissue-engineering triad are already present in revascularization procedures: stem cells, scaffold (blood clot), and growth factors (from dentin and blood). Preclinical studies that evaluated the effect of irrigants[194,300,310] and medicaments[15,248] on the survival of stem cells, release of growth factors from dentin,[270] and odontoblastic differentiation[52,106] are shaping the future generations of regenerative procedures. Furthermore, the incorporation of other scaffolds such as PRP, PRF, and gelatin sponges have been used in the clinic practice with encouraging results.[173] Discussion about the requirements of an appropriate scaffold and growth factor was begun in the 1960s by Nygaard-Østby,[221] who had no access to our contemporary instruments, materials, and the entire knowledge base of tissue engineering. Although clinical revascularization procedures do not constitute the ideal regenerative treatment, it is important to note that they do generate a scaffold (fibrin) and growth factors (from platelets and access to proteins embedded in the dentinal walls), and the clinical outcome results in continued radiographic root development of the immature permanent tooth with a diagnosis of pulpal necrosis. Thus REPs provide a treatment of high clinical value in cases with an otherwise poor prognosis. Histologic analyses of REPs conducted in patients or animals suggest that the increase in root dimensions is often due to deposition of cementum-like material, osteodentin, dentin, or bone.

Future clinical research will likely focus on further translating basic research findings into improved regenerative procedures. For example, the formation of a cementum-like material on the dentinal walls may lead to studies evaluating benefits of REPs for overall tooth resistance to fracture. In addition, it is clear that the multipotent nature of many MSC types could contribute to the finding of cementum deposition. Controlled differentiation of stem cells into odontoblasts is an important area of research and amenable to tissue-engineering concepts. The development of delivery systems that permit structural reinforcement of the cervical area (or ideally, the pulp chamber) might provide clinical opportunities to regenerate lost tooth structure, thereby permitting natural teeth to be retained instead of possible fracture or extraction. Finally, the ultimate and long-term goal for REPs should be to treat the fully formed permanent tooth. Although this situation is more complex than the immature tooth with an open apex and a ready source of stem cells, it provides the unique potential of saving the natural dentition while restoring the sensory, immunologic, and defensive properties of the pulp-dentin complex.

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